

Real word currents and native source steps Complete original-metric signatures, volumes and marked-current curvature

Split-Zero research programme

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The original arithmetic quotient has a complete positive source minimum. For a real polynomial word of degree $d \leq q/2$, its Hermitian current has d positive, d negative and $q - 2d$ zero directions. We calculate its full boundary Gram and signed product, extend the finite signature calculation to rational and real divisor words, and prove that the actual adjacent source addition preserves polynomial inertia throughout the step. The original observation retains the dimension bounds proved in WG12-13.

The same source step has an exact marked-current formula. Its curvature is determined by the change in the observed trace, equivalently the weighted imaginary displacement of the two actual dual-number collision fibres. This transports the new geometric datum into the original scalar sign test. All complete proofs and source dependencies are included. The particular terminal imaginary pairing remains unevaluated; no RH conclusion is asserted.

The first proof gives the current integrated receivers. The independent word proof supplies full sign-coordinate and denominator arguments. The complete incoming WS proof is retained with its original scope; WG and the new native-step proofs state its strengthened receivers. The published 028 inverse and constant corrections continue to govern earlier claims. Human sources and pinned programme proof links occur at use.

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Every polynomial boundary column retains its actual quotient metric

$$[p_{N-d+1}], \dots, [p_N] \mid [p_{N+1}], \dots, [p_{N+d}]$$

$$F: \mathbb{C}^{2d} \rightarrow E, \quad K_0 = F^* G_N F > 0$$

$$\langle x, W_p x \rangle_G = \|c_+\|^2 - \|c_-\|^2$$

$$\ln W_{p,t} = (d, d, q - 2d) \quad \text{for all } 0 \leq t \leq 1$$

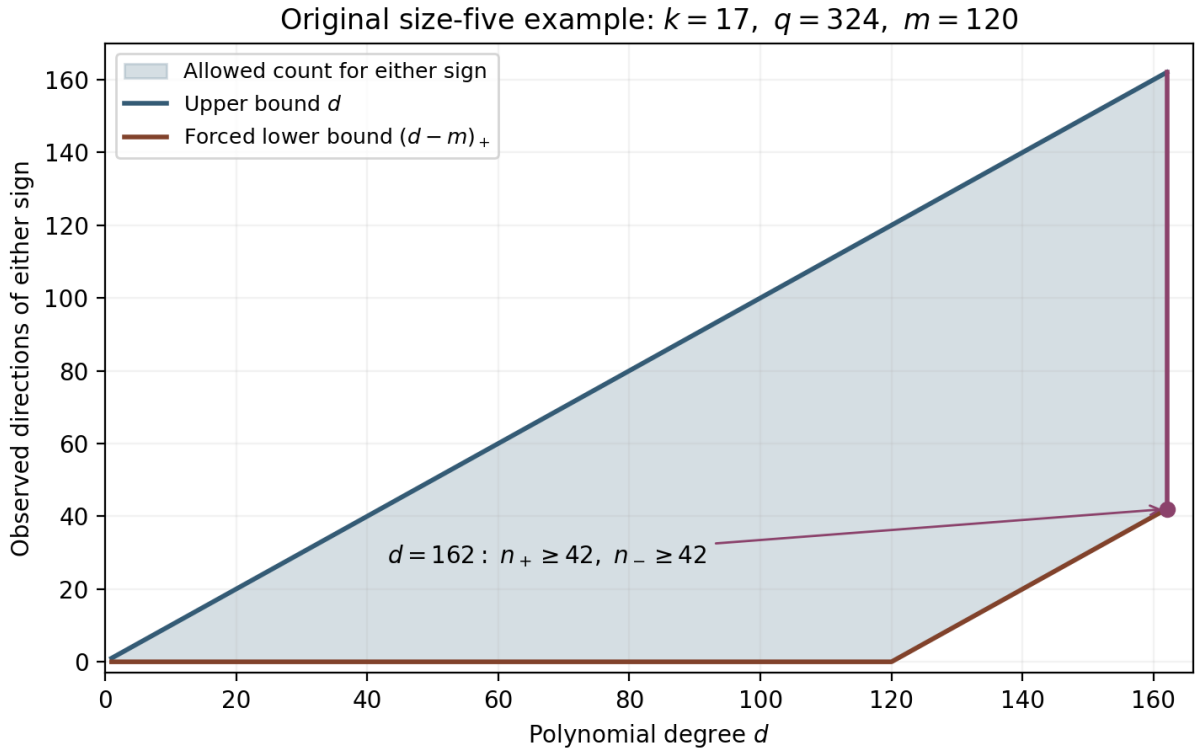


Figure 1: Upper: the full boundary and sign-coordinate maps of WD5-20 and WG5-8. The original metric K_0 remains present in the inverse coordinate map. The native adjacent-step inertia is proved in NW6-16. Lower: WG12's exact dimension bounds at the original size-five degree $k = 17$. The shaded interval is a proved range, not computed observed eigenvalue counts. At half degree, every native source-step point has at least 42 observed directions of each sign. The observation and all kernel directions retain their actual metrics.

The actual source step carries the collision trace into curvature

$$D(t) = G_N^{-1} + t u u^*, \quad u = b_{N+1} / \sqrt{\omega_{N+1}}$$

$$\chi_{\text{coll}} = a_{N+1} \text{Im} \sigma_{*, N+1} - a_N \text{Im} \sigma_{*, N}$$

$$\Delta \text{tr} W_B = 2 \chi_{\text{coll}}, \quad \lambda = \frac{(1+\beta)t}{1+\beta t}$$

$$\Phi = (1-\lambda)\Phi_N + \lambda\Phi_{N+1} + \frac{2W\chi_{\text{coll}}}{1+\beta}\lambda(1-\lambda)$$

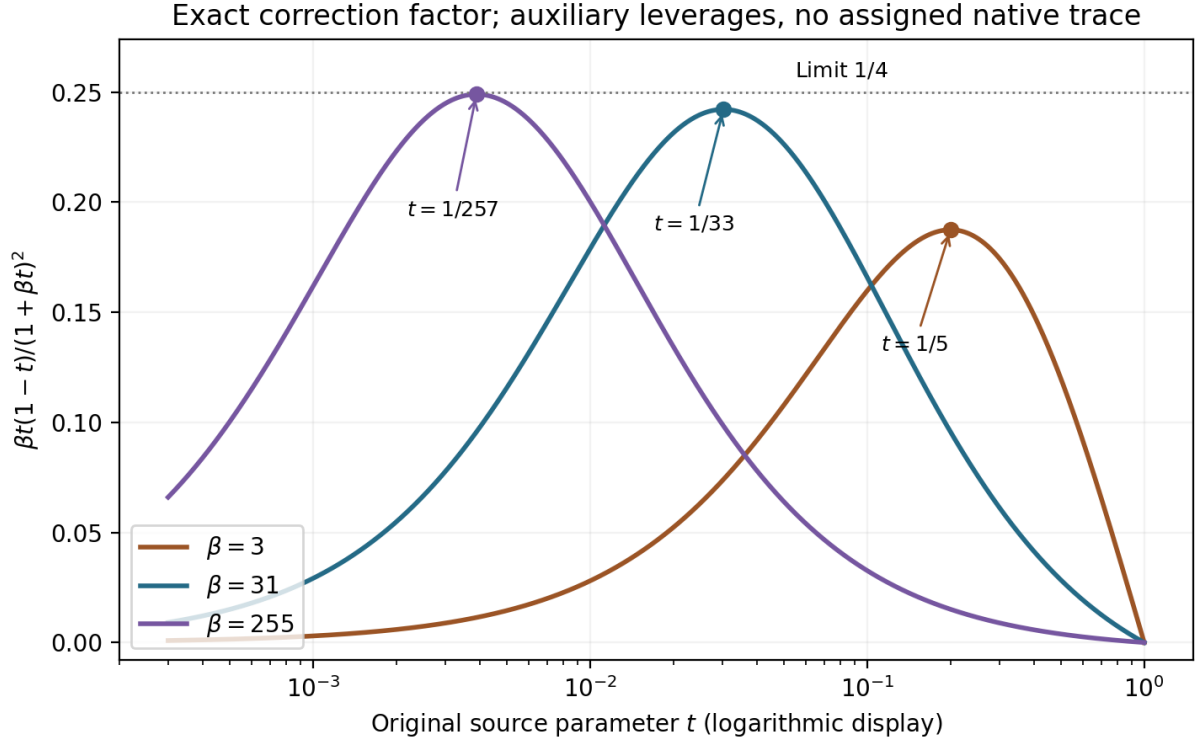


Figure 2: Upper: NV25-26 and WG14-15 carry the two collision fibres into the current's exact curvature. Each endpoint uses its own support weight. Lower: the exact scalar correction profile at three stated auxiliary leverages. Multiplication by $Y_0 \rho_f \Delta \text{tr} W_B$ gives the native correction; no trace value is assigned by the picture. Every marked peak lies at $t = 1/(\beta + 2)$ and has height $\beta/[4(1 + \beta)]$. The original $t = 0$ endpoint is omitted only from the logarithmic plotting axis; its value is exactly zero.

Whole-space words through the original source and observation

22 September 2026. WG1-WG16. This proof connects the whole-space word calculation to the actual adjacent source steps, the complete original observation and the earlier cutoff estimates. The accompanying NV1-26, NW1-18, WD1-49 and WS1-61 supply complete proofs, not assumed intermediate claims.

1. Which original objects receive the results

Use the full native polynomial Q_k , source $w_h^{*k}(y) dy$, multiplication $M[f] = [yf]$, and onto observation $\Lambda : E \rightarrow B$ of RC1-2 and OCP1-7. Here $k \geq 17$, $k \equiv 1 \pmod{4}$, $q = (k+1)^2$, $0 < \delta < 1/2$, $\gamma > 2$, and the stipulated quartet and admitted period remain fixed. The exact pairing of conjugate factors in WD43 proves $Q_k(y) > 0$ for every real y . The full minimum metric is

$$G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad q-1 \leq N \leq 2q, \quad (\text{WG1})$$

where J_N is the actual polynomial remainder map and H_N is the complete original source moment Gram. WD2-3a proves WG1, including its minimum section and all polynomial relations. The four-return operator is

$$\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}. \quad (\text{WG2})$$

The index $N+d$ used to recover a finite boundary Gram is an auxiliary exact covariance update. All currents and all four returned values are measured at their original G_N .

2. Source-degree control of the complete boundary Gram

The source comparison needed by WS30 is available at every finite degree, with explicit degree dependence. Precisely, FULL_WINDOW_SOURCE_PRODUCTS.tex, FW7 gives

$$\begin{aligned} \ell(n) &= \frac{a_k}{[1 + 4(n + M_0)^2]^{M_0}}, \quad u = A_k, \quad M_0 = 21 + 4m_0, \\ \ell(n) \|P\|_\sigma^2 &\leq \|P\|_\mu^2 \leq u \|P\|_\sigma^2, \end{aligned} \quad (\text{WG3})$$

for every polynomial of degree at most n , with $\log(A_k/a_k) = O_h(k + \log k)$. The reference measure has mass $\sqrt{2\pi}$; its monic norms are $h_j = \sqrt{2\pi} j!(1/2)_j$. The source exponent M_0 is fixed and is unrelated to the operator M .

Set $n_* = 2q + d + 1$, $\ell_* = \ell(n_*)$, $\kappa_* = u/\ell_*$. This is a substitution into WG3, not reuse of a constant that was only defined at degree $2q + 2$. For every $1 \leq d \leq q/2$,

$$\log \kappa_* = O_h(k + \log(q + d + 2)). \quad (\text{WG4})$$

The Gamma multiplication recurrence bounds multiplication on degree n by $2(n+1)$; its two off-diagonal entries have size below $n+1$, and the sum of the two shift norms proves this bound. Applying WG3 to yP and P proves $\|yP\|_\mu \leq 2(n+1)\sqrt{\kappa_*} \|P\|_\mu$ at every used degree. Applying the same argument to QP gives the corresponding bound for the relation measure $Q^2 d\mu$. In particular every zero of a used monic orthogonal polynomial is bounded by this compressed multiplication norm: the three-term recurrence identifies that polynomial with the characteristic polynomial of the real symmetric Jacobi compression. The recurrence itself follows by expanding yp_j in the orthogonal basis; symmetry removes all entries more than one degree below the diagonal. This is the classical source convention in Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt, DLMF18.2(iv).

For completeness, retain all three relation blocks of WS25:

$$R_H = \begin{pmatrix} X & Y \\ 0 & Z_0 \end{pmatrix}, \quad F = [I_d, O_d Z_0], \quad F^* G_N F = K_0 = \begin{pmatrix} I - XX^* & -Y \\ -Y^* & I - Z_0^* Z_0 \end{pmatrix} > 0. \quad (\text{WG5})$$

Here I_d, O_d are the consecutive incoming and outgoing orthonormal source classes; X has its actual, possibly smaller, width. This identity uses every source relation: the high-row Gram at $N + d$ is $I - R_H R_H^*$; its lower block and the exact update $G_{N+d}^{-1} = G_N^{-1} + O_d O_d^*$ give $(I + O_d^* G_N O_d)^{-1} = Z_0 Z_0^*$. The high-row cross block then gives $I_d^* G_N O_d = -Y Z_0^{-1}$. Substituting back into the upper block gives $I_d^* G_N I_d = I - XX^*$, and multiplying by $\text{diag}(I, Z_0)$ gives WG5. Thus the frame change and its retained metric are explicit.

The roots of Q and of all source and relation polynomials needed here lie in the bound

$$R_* = \max\{1, k\sqrt{\delta^2 + \gamma^2}, 2(N + d + 1)\sqrt{\kappa_*}\}, \quad B_* = 2(N + d)R_*.$$

For a monic degree- n polynomial with roots of modulus at most R_* , its coefficient h positions below the top has modulus at most $(nR_*)^h$, by the elementary symmetric sums. Expanding the monic relation QU_{n-q} in the monic source basis and eliminating successive leading coefficients therefore bounds its expansion coefficient h positions below the top by $(2nR_*)^h$: induction gives the coefficient sum bound $1 + \sum_{j=1}^{h-1} 2^j \leq 2^h$. For the relevant indices $i \leq n$, $n - i \leq 2d$, their normalized coefficient has the additional exact factor

$$\sqrt{\omega_i/\omega_n} \sqrt{\omega_n/\nu_{n-q}}.$$

Minimizing WG3 over all monic polynomials gives $\ell_* h_i \leq \omega_i \leq u h_i$. The high indices satisfy $h_i \leq h_n$, since each ratio $h_{j+1}/h_j = (j+1)(j+1/2)$ exceeds one there. Consequently

$$\max(\|X\|, \|Y\|, \|Z_0\|) \leq \delta_* := d\sqrt{\kappa_*} B_*^{2d} \quad (\text{WG6})$$

$$\max_{\max(q, N-d+1) \leq n \leq N+d} \sqrt{\omega_n/\nu_{n-q}}.$$

No off-diagonal relation coefficient has been omitted. The factor d bounds the matrix norm by its maximum entry times the square root of its actual row/column counts.

The full-root monic theorem RC23, using LRS11 holds for $0 \leq j \leq 2q$. Every $j = n - q$ in WG6 lies in this interval, since $n \leq 2q + d$ and $d \leq q/2$. Its comparison with WG3 and its logarithmic endpoint derivative estimate give

$$\log \delta_* \leq -q\psi((N + 1 - q)/q) + O_h(k \log(q + 2) + d[k + \log(q + 2)]). \quad (\text{WG7})$$

Indeed the argument shift is at most $(d+1)/q$; integrating $1 + |\log t|$ near zero bounds its q -weighted contribution by $O((d+1) \log(q+2))$. All remaining factors are the explicit terms of WG6 and WG4. For $d = o(k)$, WG7 tends to a negative constant times q , uniformly across the original source window. Put $\rho_* = \delta_* + \delta_*^2$; when $\rho_* < 1$, WG5 has $(1 - \rho_*)I \preceq K_0 \preceq (1 + \rho_*)I$. This is an effective finite guard with the actual scalar norms in WG6, and its eventual validity has just been proved.

3. Signed volume and the four original cutoffs

For a real polynomial p of degree $d \leq q/2$, let $c_d \neq 0$ be its leading coefficient and let B_p be the full triangular outgoing matrix of WD6-7. In WG5 put $D_p = Z_0^{-1} B_p$. The complete current is

$$W_p = F \begin{pmatrix} 0 & -iD_p^* \\ iD_p & 0 \end{pmatrix} F^{\dagger G_N}.$$

The isometry $FK_0^{-1/2}$ converts its nonzero part to the congruence of the displayed middle matrix by $K_0^{1/2}$. Its determinant is therefore the middle determinant times $\det K_0$. The two triangular diagonals are

$$(B_p)_{ss} = c_d \sqrt{\omega_{N+s}/\omega_{N-d+s}},$$

$$(Z_0)_{ss} = \sqrt{\omega_{N+s}/\nu_{N+s-q}}.$$

They cancel only the indicated intermediate norms, proving the exact identity

$$\text{pdet } W_p = (-1)^d |c_d|^{2d}$$

$$\det K_0 \prod_{s=1}^d \frac{\nu_{N+s-q}}{\omega_{N-d+s}}. \quad (\text{WG8})$$

Here pdet is the product of every nonzero eigenvalue, with sign. Original coefficient conjugation sends W_p to $-W_p$, so those eigenvalues pair as $\pm\sigma_j$, $\sigma_j > 0$. Every factor in WG8 is independent of the lower coefficients of p . Thus the full signed product is exactly unchanged under every real change of lower coefficients with fixed c_d , even unbounded changes.

Use WG3 for both monic norms in WG8. The relation comparison uses all $Q^2 d\mu$; its Gamma ratio contributes $2q\psi((N+s-q)/q)$, and the remaining factorial ratio h_{N+s}/h_{N-d+s} has logarithm $O(d\log(q+2))$. Summing d factors, using the endpoint shift bound from WG7 and $|\log \det K_0| \leq 2d \max\{-\log(1-\rho_*), \log(1+\rho_*)\}$, proves, for $d = o(k)$,

$$\sum_{j=1}^d \log \sigma_j = d \log |c_d| + dq\psi((N+1-q)/q)$$

$$+ O_h(dk \log(q+2) + d^2 \log(q+2)). \quad (\text{WG9})$$

Applying WG2 at the exact parameters $0, 1/q, 1, 1+1/q$ gives

$$\mathcal{R} \sum_{j=1}^d \log \sigma_j = 2d[\psi(0) - \psi(1)]q$$

$$+ O_h(dk \log(q+2) + d^2 \log(q+2)). \quad (\text{WG10})$$

Its error divided by dq tends to zero. The leading coefficient must be the same word at all four cutoffs; then its logarithm cancels exactly. The original kernel-return constant remains $C_\partial = 2J(1)$; it is a different defined functional of the same ψ , connected by the proved identity

$$C_\partial = 8 \int_0^1 \psi(s) ds + 4\psi(0) - 8\psi(1).$$

This is the published DF15-16 correction. The new outward integer enclosure in WS61 strengthens its certificate; it does not reopen the sealed027 source. The exact whole-space formulas WG5 and WG8 remain valid at all $d \leq q/2$, without extending the asymptotic small- d guard beyond its proof.

4. Native source activation and the full observed image

NW1-18 proves the following new statement using the actual source column:

$$D(t) = G_N^{-1} + t b_{N+1} b_{N+1}^\top / \omega_{N+1},$$

$$\text{In } W_{p,t} = (d, d, q-2d) \quad (0 \leq t \leq 1, 1 \leq d \leq q/2). \quad (\text{WG11})$$

For $2d < q$ its explicit congruence eliminates the last row/column by the exact skew identity $a^\top B_N^{-1} a = 0$. For $2d = q$, its Pfaffian is the affine combination of the two nonzero, same-sign endpoint Pfaffians; the positive determinant orientation of every q -column consecutive source frame is proved there by an explicit measure pushforward. These arguments retain the moving radical, including the entire original kernel dimension.

Let $m = \dim \ker \Lambda$. For each t , put

$$Q_B(t) = (\Lambda D(t) \Lambda^*)^{-1}, \quad J_t = D(t) \Lambda^* Q_B(t)^{1/2}.$$

Then $J_t^* G(t) J_t = I_B$. If $H_+(t)$ is any positive d -dimensional subspace of the full current, the dimension formula gives $\dim(H_+(t) \cap \text{ran } J_t) \geq d - m$. The isometry pulls this intersection to a positive subspace of the actual observed current. Conversely any positive observed subspace maps into a positive full subspace of dimension at most d . The same argument for negative subspaces proves uniformly throughout the native step

$$(d - m)_+ \leq n_\pm(J_t^\dagger W_{p,t} J_t) \leq d. \quad (\text{WG12})$$

At $d = q/2$, its nullity is at most m , because $\dim B = q - m$ and the two lower bounds sum to $q - 2m$.

On the actual size-five orbit domain, the exact invariant restriction theorem OQX3 gives $m = 8k - 16$. Hence every half-degree native step retains at least

$$\frac{q}{2} - m = \frac{(k - 3)(k - 11)}{2} \quad (\text{WG13})$$

observed directions of each sign. The dimension theorem's size-five domain is retained; for any other observation WG12 uses its actual m . This proves a quantitative obstruction to a positive semidefinite observed half-degree current throughout the original source step, and identifies the positive and negative observed spaces through J_t . It does not select a sign for a prescribed terminal vector.

5. The collision data now determine source curvature

Use the original adjacent paths $N = q - 1 \rightarrow q$ and $N = 2q - 1 \rightarrow 2q$, separately. Write Φ_i for the marked current at an endpoint, $\tau_i = \text{tr } W_{B,i}$, and retain each endpoint's own observed source columns u_i, v_i , $a_i = \|u_i\|^2$, $r_i = u_i^* v_i$. The complete area estimate gives $a_i d_i - |r_i|^2 > 0$.

For $F_i(\sigma) = (\epsilon_i v_i + \sigma u_i) u_i^*$, multiplication gives $F_i^2 = (a_i \sigma + \epsilon_i r_i) F_i$. At $\sigma_{*,i} = -\epsilon_i r_i / a_i$, the generator is $\epsilon_i \sqrt{a_i d_i - |r_i|^2} h_i g_i^*$ in the actual current-plane isometry. It is nonzero square-zero. The matrices I, F_i are linearly independent: a scalar multiple of I that is nilpotent must be zero, whereas this generator is nonzero. The map from the dual-number algebra sending its generator to F_i is consequently injective and onto this two-dimensional algebra. Tracing the rank-one boundary term gives $\tau_i = -2\epsilon_i \Im r_i = 2a_i \Im \sigma_{*,i}$, retaining the positive real ϵ_i .

For the native source column $u = b_{N+1} / \sqrt{\omega_{N+1}}$, put $z = \Lambda u$, $\beta = z^* Q_B(0) z$, $w = |z^* Q_B(0) \xi|^2$, and

$$\chi_{\text{coll}} = a_{N+1} \Im \sigma_{*,N+1} - a_N \Im \sigma_{*,N}.$$

The exact minimum-map calculation NV1-9 gives

$$\Phi(t) = \frac{(1-t)\Phi_N + t(1+\beta)\Phi_{N+1}}{1+\beta t} + \frac{2t(1-t)w \chi_{\text{coll}}}{(1+\beta t)^2}. \quad (\text{WG14})$$

Thus the new collision datum supplies the curvature coefficient of the actual marked source path. With the explicitly invertible coordinate $\lambda = (1 + \beta)t/(1 + \beta t)$,

$$\boxed{\begin{aligned} \frac{d^2\Phi}{d\lambda^2} &= -\frac{4w}{1+\beta}\chi_{\text{coll}}, \\ t_{\text{peak}} &= \frac{1}{\beta+2}, \quad \Phi(t_{\text{peak}}) - \frac{\Phi_N + \Phi_{N+1}}{2} \\ &= \frac{w}{2(1+\beta)}\chi_{\text{coll}}. \end{aligned}} \quad (\text{WG15})$$

The support weights cannot be suppressed: the two endpoints generally have different a_i and different source metrics. NV16-20 carries the earlier doubled-rate terminal support estimate into the factor $w/(1 + \beta)$, with its full finite error.

The exact whole-step sign criterion follows by putting $\kappa = 2w\chi_{\text{coll}}/(1 + \beta)$. For non-negative endpoint values $A = \Phi_N, B = \Phi_{N+1}$, the path is nonnegative precisely when

$$\boxed{\kappa \geq -(\sqrt{A} + \sqrt{B})^2.} \quad (\text{WG16})$$

Indeed division of $(1 - \lambda)A + \lambda B + \kappa\lambda(1 - \lambda)$ by $\lambda(1 - \lambda)$ reduces the question to the infimum of $A/\lambda + B/(1 - \lambda)$. Subtracting $(\sqrt{A} + \sqrt{B})^2$ yields $[\sqrt{A}(1 - \lambda) - \sqrt{B}\lambda]^2/[\lambda(1 - \lambda)]$. The zero-endpoint limits give the same criterion. NV21-24 also proves the complete negative cone and its boundary source coordinate. Actual endpoint values and weighted collision displacements remain the precise scalar data to calculate; no value is supplied by the sign count alone.

6. Earlier results now receiving the stronger calculation

The finite WS1-24 statements are supported by the independent complete WD1-49 derivation, including all rational denominator maps, repeated factors, isotropic extensions and high-degree divisor congruences. WS25-33 receives the degree-explicit comparison WG3-10. The mixed quartic construction WS36-38 retains the full arithmetic mixed maps $C = P_K M|_{K^\perp}$, $D = P_B M|_K$. Its native rank bound uses both the size-five dimension above and the original nonzero four-corner conductor condition; the complete coefficient proof of $\dim K_8 \leq 128$ is retained at the bounded-observation source, equations 10-13. Its phase orbit keeps the observation but has the exact action defect $MU_\theta - U_\theta M = (e^{i\theta} - 1)(PMP_K - P_KMP)$; it therefore supplies whole-fibre sign witnesses and the map recording their change of arithmetic class.

The general source pencil WS39-44 is retained with all singular blocks. NW11-16 and WG11 prove the stronger constant-inertia result for the actual adjacent polynomial source; WG12-13 transports its forced signs through the moving original quotient. The latest EE1-48 already proves the intrinsic and measured inverses on the longer stated clock, as well as all fixed $b > 0$. WS46-54 supplies an additional direct full-characteristic-polynomial contour derivation; its narrower description of what remains open is superseded by EE, not promoted into a new missing task.

The unfinished signed scalar is still the actual imaginary pairing of the retained terminal class, now joined by exact endpoint/trace and collision receivers WG14-16. The next calculation should evaluate those actual coefficient-dependent endpoint traces or pairings, with the complete source minimum, rather than repeat the already established whole-space signatures or recovery maps.

Canonical polynomial, rational, and complete real-divisor currents

22 September 2026. Independent derivation of incoming WS1-WS24, including WS21a-e. The input is Whole_Space_Signs_20260922/COMPLETE_PROOFS.md. The proof below establishes the asserted signatures without relying on the input's finite-check count. No mathematical defect in those signatures was found. The maximal-isotropic extension argument and the large-degree divisor congruence are supplied explicitly.

1. The unchanged quotient and its attained metric

Let μ be the original positive measure on $\mathbb{R}$, with all moments used below finite and with positive definite polynomial Gram matrices. Let $Q \in \mathbb{R}[y]$ be monic of degree q , and suppose $Q > 0$ almost everywhere for μ . Set $\mathcal{P}_j = \{0\}$ for $j < 0$, and otherwise let $\mathcal{P}_j$ be the complex polynomials of degree at most j . Every source inner product is conjugate-linear in its first argument:

$$\langle f, g \rangle_\mu = \int \overline{f(y)} g(y) d\mu(y). \quad (\text{WD1})$$

For $N \geq q - 1$, let $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[y]/(Q)$ be the remainder map. Its kernel is exactly $Q\mathcal{P}_{N-q}$. Define $H_N = (Q\mathcal{P}_{N-q})^\perp \subset \mathcal{P}_N$. The restriction $J_N|_{H_N}$ is bijective: it is injective because a vector lying in a subspace and its orthogonal complement has zero norm, and it is onto by subtracting the orthogonal projection onto the kernel from any representative. Define

$$R_N = (J_N|_{H_N})^{-1}, \quad \langle x, z \rangle_{G_N} = \langle R_N x, R_N z \rangle_\mu. \quad (\text{WD2})$$

Every other representative is $R_N x + v$, $v \in \ker J_N$, and has squared norm $\|R_N x\|_\mu^2 + \|v\|_\mu^2$. This proves that WD2 is the complete attained quotient minimum. Moreover,

$$J_N R_N = I_E, \quad R_N J_N = P_{H_N}, \quad J_N^\dagger = R_N, \quad R_N^\dagger R_N = I_E. \quad (\text{WD3})$$

Indeed $\langle J_N f, x \rangle_{G_N} = \langle P_{H_N} f, R_N x \rangle_\mu = \langle f, R_N x \rangle_\mu$. In the following formulas $G = G_N$, and an adjoint always uses this metric on E and the stated source metric or ordinary Euclidean metric on the other space.

In coefficient coordinates let H_N be the full positive source moment matrix and write J_N also for the matrix of the remainder map. Then the complete formulas are

$$G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad R_N = H_N^{-1} J_N^* G_N. \quad (\text{WD3a})$$

The covariance is positive definite because J_N^* is injective and $H_N^{-1} > 0$. These displayed matrices satisfy $J_N R_N = I$, and for $v \in \ker J_N$, $v^* H_N R_N x = (J_N v)^* G_N x = 0$. They therefore give precisely the inverse restriction in WD2. Multiplication also gives $R_N^* H_N R_N = G_N$, verifying the full metric formula directly.

Let $M : E \rightarrow E$ be $M[f] = [yf]$. For any polynomial or defined rational operator T , put

$$W_T = i(T - T^{\dagger G}), \quad \mathfrak{W}_T = G W_T = i(GT - T^* G). \quad (\text{WD4})$$

The Hermitian operator $G^{1/2} W_T G^{-1/2}$ and the Hermitian form matrix $\mathfrak{W}_T$ have the same inertia, because one is the congruence of the other by $G^{-1/2}$. Thus operator and form signatures refer to the same positive, negative, and zero counts.

These statements retain the source mass. Let p_j be the unique real monic polynomial orthogonal to $\mathcal{P}_{j-1}$, let $\omega_j = \int |p_j|^2 d\mu > 0$, and put $\phi_j = p_j / \sqrt{\omega_j}$. They exist by orthogonal projection of y^j onto $\mathcal{P}_{j-1}$; reality follows from the real moment matrix. In particular $\omega_0 = \mu(\mathbb{R})$. No rescaling of μ has been made.

2. The complete polynomial boundary factor

Let $p(y) = \sum_{j=0}^d a_j y^j \in \mathbb{R}[y]$, with $a_d \neq 0$ and $1 \leq d \leq q/2$. Define actual quotient-column maps from $\mathbb{C}^d$ to E by

$$I_d = [J_N \phi_{N-d+1}, \dots, J_N \phi_N], \quad O_d = [[\phi_{N+1}], \dots, [\phi_{N+d}]]. \quad (\text{WD5})$$

The outgoing polynomial may have degree greater than N , but its class is measured by exactly G_N ; its representative is $R_N[\phi_j]$. Define

$$B_p[r, s] = \langle \phi_{N+r}, p \phi_{N-d+s} \rangle_\mu, \quad 1 \leq r, s \leq d. \quad (\text{WD6})$$

For $r > s$, the second polynomial has degree at most $N + s < N + r$, so $B_p[r, s] = 0$. When $r = s$, subtracting $a_d \sqrt{\omega_{N+s}/\omega_{N-d+s}} \phi_{N+s}$ from $p \phi_{N-d+s}$ removes its leading coefficient. Its remaining degree is less than $N + s$, proving

$$B_p[s, s] = a_d \sqrt{\frac{\omega_{N+s}}{\omega_{N-d+s}}} \neq 0. \quad (\text{WD7})$$

Thus B_p is invertible, with its full source-norm ratios.

For $x \in E$, expand $R_N x$ in the orthonormal source basis. Its last d coefficients are $I_d^\dagger x$, because WD3 gives $\langle J_N \phi_j, x \rangle_G = \langle \phi_j, R_N x \rangle_\mu$. Multiplication by p sends every lower coefficient into $\mathcal{P}_N$. The outgoing coefficients of $p R_N x$ are consequently exactly $B_p I_d^\dagger x$. The part in $\mathcal{P}_N$, followed by the remainder map, is

$$C_p = R_N^\dagger P_{\mathcal{P}_N} p(y) R_N.$$

For $x, z \in E$, its pairing is $\langle R_N x, p R_N z \rangle_\mu$. Since p is real, this is the conjugate of the reversed pairing, so $C_p = C_p^\dagger$. Taking quotient classes of the full expansion therefore proves

$$p(M) = C_p + O_d B_p I_d^\dagger. \quad (\text{WD8})$$

Set

$$Z_p = [I_d, O_d B_p], \quad J_d = \begin{pmatrix} 0 & -i I_d \\ i I_d & 0 \end{pmatrix}.$$

Subtracting the adjoint of WD8, including the reversed matrix order, gives

$$W_p = Z_p J_d Z_p^\dagger. \quad (\text{WD9})$$

3. Positivity proves all boundary columns independent

Suppose a linear combination F of the $2d$ consecutive source polynomials in WD5 has zero quotient class. Polynomial divisibility gives $F = Qh$, and

$$\deg h \leq N + d - q \leq N - d. \quad (\text{WD10})$$

Every polynomial appearing in F is orthogonal to $\mathcal{P}_{N-d}$. Therefore

$$0 = \langle h, F \rangle_\mu = \int Q(y) |h(y)|^2 d\mu(y). \quad (\text{WD11})$$

The integrand is nonnegative and vanishes in integral only if $h = 0$ almost everywhere, since $Q > 0$ almost everywhere. Positive definiteness of the source Gram then gives $h = 0$ as a polynomial. Hence $F = 0$, and linear independence of the orthogonal source polynomials makes all its coefficients zero. This proves that $[I_d, O_d]$ and Z_p are injective. No root interpolation, root separation, or factor coprimality has entered the proof.

Let

$$R_p = Z_p^* G Z_p > 0, \quad V_p = Z_p R_p^{-1/2}. \quad (\text{WD12})$$

Then $V_p^\dagger V_p = I_{2d}$. Its range is $\text{ran } Z_p$, and

$$V_p^\dagger W_p V_p = R_p^{1/2} J_d R_p^{1/2}. \quad (\text{WD13})$$

On $(\text{ran } Z_p)^\perp = \ker Z_p^\dagger$, WD9 is zero. On the range, WD13 is nonsingular. The eigenvalues of J_d are $+1$ and -1 , each d times: vectors (u, iu) and $(u, -iu)$ give the respective eigenspaces. Invertible congruence preserves the maximal dimensions of positive and negative subspaces, since it maps such subspaces bijectively. Consequently

$$\boxed{\text{In } W_p = (d, d, q - 2d).} \quad (\text{WD14})$$

The exact relation with the original first-order current is

$$W_p = \sum_{j=1}^d a_j \sum_{h=0}^{j-1} (M^\dagger)^h W_y M^{j-1-h}. \quad (\text{WD15})$$

For each j , insert $W_y = i(M - M^\dagger)$; every internal term cancels the following term and the two endpoints are $i[M^j - (M^\dagger)^j]$. This proves WD15 without commuting M with its adjoint. For the unchanged physical action $\mathcal{A} = kI/2 + iM$,

$$\mathcal{A}^* G + G \mathcal{A} - kG = G W_y. \quad (\text{WD16})$$

4. All sign coordinates, zero directions, and isotropic spaces

For an arbitrary $x \in E$, write $t = Z_p^\dagger x = (a, b)$. The full inverse-coordinate decomposition is

$$x = Z_p R_p^{-1} t + x_0, \quad x_0 \in \ker Z_p^\dagger. \quad (\text{WD17})$$

It is unique because applying $Z_p^\dagger$ to its first summand gives t , and the two summands lie in orthogonal subspaces. In particular every t occurs and

$$\|x\|_G^2 = t^* R_p^{-1} t + \|x_0\|_G^2. \quad (\text{WD18})$$

Set $c_+ = (a - ib)/\sqrt{2}$, $c_- = (a + ib)/\sqrt{2}$. Matrix multiplication gives

$$\langle x, W_p x \rangle_G = 2 \text{Im}(a^* b) = \|c_+\|^2 - \|c_-\|^2. \quad (\text{WD19})$$

Positive, negative, and zero form values are precisely the three corresponding norm comparisons. The radical is $a = b = 0$; zero form value alone does not imply membership in the radical.

Put

$$S_d = \frac{1}{\sqrt{2}} \begin{pmatrix} I_d & I_d \\ iI_d & -iI_d \end{pmatrix}.$$

Then $t = S_d(c_+, c_-)$ and $S_d^* J_d S_d = \text{diag}(I_d, -I_d)$. Every maximal totally isotropic complex subspace has the exact form

$$\left\{ Z_p R_p^{-1} S_d(u, Uu) + x_0 : u \in \mathbb{C}^d, x_0 \in \ker Z_p^\dagger \right\}, \quad (\text{WD20})$$

$$U^* U = I_d,$$

and has dimension $q - d$. Here maximality can mean maximal by inclusion. To prove this, a maximal isotropic subspace contains the radical, because adjoining a radical vector

preserves every pairing. Quotient by the radical and use the (c_+, c_-) coordinates. Projection of an isotropic space onto the positive coordinates is injective: a vector $(0, v)$ has value $-\|v\|^2$. Thus the space is the graph of an isometry $U_0 : H_+ \rightarrow H_-$ between subspaces of $\mathbb{C}^d$. If their common dimension is less than d , choose unit vectors perpendicular to H_+ and H_- , and map one to the other. This extends the graph to a strictly larger isotropic space, a contradiction. Hence both subspaces are the full $\mathbb{C}^d$, and U_0 is unitary. Conversely a unitary graph is isotropic and already has the maximum possible dimension. WD17 restores every original coordinate and metric.

Since Q and the moments are real, coefficient conjugation preserves the minimum problem and is antiunitary in G . For real p , it sends W_p to $-W_p$. Applying conjugation to an eigenvector proves exact eigenvalue pairing $+\sigma_j, -\sigma_j$, including multiplicities.

5. The reduced rational word and repeated poles

Let $r = p/s$, where $p, s \in \mathbb{R}[y]$, $s \neq 0$, $\gcd(p, s) = 1$, and $\gcd(s, Q) = 1$. Let $d = \max(\deg p, \deg s)$ and assume $1 \leq d \leq q/2$. The Euclidean identity gives polynomials u, v with $us + vQ = 1$, so $s(M)^{-1} = u(M)$. This is an inverse on the entire original quotient. Conversely, a common root of s, Q makes $s(M)$ singular by evaluation at that root, so the invertibility condition is exact. Set

$$T_r = p(M)s(M)^{-1}. \quad (\text{WD21})$$

No rational function is integrated across a real pole. All integrations below are polynomial source pairings already defined in WD1.

The numerator $s(x)p(y) - p(x)s(y)$ vanishes at $y = x$. Its quotient by $y - x$ is symmetric in x, y , has real coefficients, and has degree at most $d - 1$ in each variable. Define its real symmetric coefficient matrix $\mathcal{B}$ by

$$\frac{s(x)p(y) - p(x)s(y)}{y - x} = \sum_{i,j=0}^{d-1} \mathcal{B}_{ij} x^i y^j. \quad (\text{WD22})$$

We prove invertibility without assuming that any root is simple. If necessary interchange s, p , which changes $\mathcal{B}$ to $-\mathcal{B}$, to arrange $\deg s = d$. Write its leading coefficient as $s_d \neq 0$. In $A = \mathbb{C}[x]/(s)$, multiplication by p is invertible by the Euclidean identity. Put

$$c_s(x, y) = \frac{s(y) - s(x)}{y - x}.$$

Its d -by- d coefficient matrix is zero for $i + j > d - 1$ and is s_d for $i + j = d - 1$, so its determinant is $(-1)^{d(d-1)/2} s_d^d \neq 0$. Modulo $s(x)$, WD22 equals $-p(x)c_s(x, y)$. To verify this even in the algebra with nilpotents, multiply both sides by $y - x$. Their products agree, and multiplication by the monic polynomial $y - x$ in $A[y]$ is injective: the top nonzero coefficient of any polynomial remains its top nonzero coefficient one degree higher. Both sides have their unique remainder of degree less than d in x . Thus $\mathcal{B}$ is minus the product of two invertible coefficient matrices, and is invertible.

Retain the exact degree-one boundary factor $W_y = Z_1 J_1 Z_1^\dagger$ from WD9, and define

$$\mathcal{Z}_d = [Z_1, M^\dagger Z_1, \dots, (M^\dagger)^{d-1} Z_1]. \quad (\text{WD23})$$

Let R_d denote the matrix with ones on its anti-diagonal and zeros elsewhere. WD15 for y^d says

$$W_{y^d} = \mathcal{Z}_d (R_d \otimes J_1) \mathcal{Z}_d^\dagger. \quad (\text{WD24})$$

By WD14 the left side has rank $2d$. The right side has rank at most $\text{rank } \mathcal{Z}_d \leq 2d$. Hence $\mathcal{Z}_d$ is injective.

Write $S = s(M)$, $P = p(M)$. Since P, S commute, $T_r S = P$. Consequently

$$S^* \mathfrak{W}_r S = i(S^* G P - P^* G S). \quad (\text{WD25})$$

To evaluate the right side, use the linear substitution $x^i y^j \mapsto (M^*)^i G M^j$ in $(y - x)$ times WD22. The image of a summand is

$$i \mathcal{B}_{ij} (M^*)^i (G M - M^* G) M^j = \mathcal{B}_{ij} (M^*)^i G W_y M^j.$$

This is a calculation with commuting left and right multiplication operations on a middle matrix; it does not commute M and G . Substitution of WD23 yields the exact denominator congruence

$$\boxed{S^* \mathfrak{W}_r S = G Z_d (\mathcal{B} \otimes J_1) Z_d^\dagger.} \quad (\text{WD26})$$

Equivalently, define the original-space map

$$Z_r = (S^{\dagger G})^{-1} Z_d.$$

Then

$$W_r = Z_r (\mathcal{B} \otimes J_1) Z_r^\dagger, \quad \text{rank } Z_r = 2d. \quad (\text{WD27})$$

Diagonalizing the real symmetric invertible matrix $\mathcal{B}$ shows that each of its d nonzero eigenvalues contributes one positive and one negative eigenvalue when tensored with J_1 . Repeating the isometry argument WD12-14 proves

$$\boxed{\text{In } W_r = (d, d, q - 2d).} \quad (\text{WD28})$$

For a constant real rational word the current is zero. A common factor in an unreduced pair must be cancelled first, with its full multiplicity. Cancellation lowers the rational degree by its actual degree; an arbitrary unreduced degree has no signature significance. Repeated roots of the reduced denominator and repeated primaries of Q have been retained throughout.

For complete rational sign coordinates, put $H_r = \mathcal{B} \otimes J_1$, $R_r = Z_r^* G Z_r > 0$. Diagonalize $\mathcal{B} = U \text{diag}(\beta_j) U^T$ with real orthogonal U . The unitary eigenvectors $(1, i)/\sqrt{2}$, $(1, -i)/\sqrt{2}$ diagonalize J_1 . Form their tensor products with the columns of U , multiply each vector by $|\beta_j|^{-1/2}$, and order the resulting positive vectors first and negative vectors second. Let the resulting invertible matrix be A_r . Then $A_r^* H_r A_r = \text{diag}(I_d, -I_d)$. For every original vector,

$$x = Z_r R_r^{-1} A_r (c_+, c_-) + x_0, \quad x_0 \in \ker Z_r^\dagger, \quad (\text{WD29})$$

and

$$\begin{aligned} \langle x, W_r x \rangle_G &= \|c_+\|^2 - \|c_-\|^2, \\ \|x\|_G^2 &= (c_+, c_-)^* A_r^* R_r^{-1} A_r (c_+, c_-) + \|x_0\|_G^2. \end{aligned} \quad (\text{WD30})$$

The same unitary-graph argument as WD20 gives all maximal isotropic subspaces. Reality of T_r gives the same exact eigenvalue pairing by the original conjugation.

For any invertible T , direct expansion gives

$$i(T^{-1} - (T^{-1})^\dagger) = -(T^{-1})^\dagger i(T - T^\dagger) T^{-1}. \quad (\text{WD31})$$

Taking $T = M^d$ gives the stated exact map between positive and negative power currents whenever $Q(0) \neq 0$.

At $q = 2d$, Z_d is square invertible. The determinants of $R_d \otimes J_1$ and $\mathcal{B} \otimes J_1$ are respectively $(-1)^d$ and $(-1)^d (\det \mathcal{B})^2$. Taking determinants in WD24 and WD26 and cancelling the common nonzero column-Gram determinant proves

$$\boxed{\det W_r = \frac{(\det \mathcal{B})^2}{|\det s(M)|^2} \det W_{y^d}.} \quad (\text{WD32})$$

For a monic $Q = \prod_{\lambda} (y - \lambda)^{m_{\lambda}}$, the exact primary decomposition of its quotient has blocks $\mathbb{C}[\eta]/(\eta^{m_{\lambda}})$. On each block $s(M)$ is triangular with diagonal $s(\lambda)$. Therefore

$$\det s(M) = \prod_{\lambda} s(\lambda)^{m_{\lambda}} = \text{Res}(Q, s). \quad (\text{WD33})$$

The Chinese-remainder decomposition is only across the distinct primary ideals, and the nilpotent powers within each block remain present. WD32 is independent of the source cutoff except through $\det W_{y^d}$. A logarithmic return with weights summing to zero consequently cancels the displayed fixed rational factor exactly.

6. Real divisors at every degree

Let $Q = DO$, where $D, O \in \mathbb{R}[y]$ are monic, and put $a = \deg D$, $r = \deg O$, so $a + r = q$. No assumption that D, O are coprime is made. Set $T = D(M)$. Cancellation in the polynomial integral domain gives

$$\ker T = \{[Ob] : \deg b < a\}, \quad \text{im } T = \{[Db] : \deg b < r\}. \quad (\text{WD34})$$

For the kernel, $Q \mid Df$ is equivalent to $O \mid f$. The classes displayed have unique representatives of degree below q , so their dimension is a . For the image, divide any polynomial f by O ; the O -multiple disappears after multiplication by D , and the displayed r basis classes are independent by their degrees. Thus $\text{rank } T = r$.

If $a = 0$, then $T = I$ and its current is zero. If $r = 0$, then $T = 0$ and its current is zero. If $1 \leq a \leq r$, WD14 applies and gives $(a, a, q - 2a)$. It remains to prove the case $1 \leq r \leq a$.

Suppose $x = [Db] \in \text{im } T \cap (\ker T)^{\perp_G}$, with $\deg b < r$. Its complete minimum representative is

$$F = R_N x = Db + Qh = Dg, \quad g = b + Oh, \quad \deg g \leq N - a. \quad (\text{WD35})$$

It is orthogonal to $Q\mathcal{P}_{N-q}$. It is also orthogonal in the source to OP_{a-1} : for such f , WD3 gives $\langle f, F \rangle_{\mu} = \langle J_N f, x \rangle_G = 0$. Division by the monic D proves the exact source-subspace identity

$$OP_{a-1} + Q\mathcal{P}_{N-q} = OP_{N-r}. \quad (\text{WD36})$$

Indeed any $u \in \mathcal{P}_{N-r}$ has division $u = Dv + w$, where $\deg w < a$ and $\deg v \leq N - q$, while all polynomials on the left have degree at most N . Because $a \geq r$, WD35 gives $\deg g \leq N - a \leq N - r$. Thus Og is one of the source polynomials in WD36, and

$$0 = \langle Og, Dg \rangle_{\mu} = \int Q(y) |g(y)|^2 d\mu(y). \quad (\text{WD37})$$

Reality of both factors has been used in this exact identity. Positivity, as in WD11, gives $g = 0$, and hence $x = 0$. We have proved

$$\text{im } T \cap (\ker T)^{\perp_G} = \{0\}. \quad (\text{WD38})$$

Write $K = \ker T$ and $H = K^{\perp_G}$, with dimensions a and r . Relative to this exact orthogonal decomposition,

$$T = \begin{pmatrix} 0 & B \\ 0 & A \end{pmatrix}, \quad W_T = \begin{pmatrix} 0 & iB \\ -iB^{\dagger} & i(A - A^{\dagger}) \end{pmatrix}. \quad (\text{WD39})$$

Here $B : H \rightarrow K$ and $A : H \rightarrow H$. If $Bh = 0$, then $Th = Ah \in H \cap \text{im } T$, so WD38 gives $Th = 0$, and then $h \in H \cap K = 0$. Hence B is injective. Set $K_1 = \text{ran } B$, $K_0 = K \ominus K_1$. Every vector of K_0 is annihilated by WD39, and $\dim K_0 = a - r = q - 2r$.

On $K_1 \oplus H$, put $C = iB : H \rightarrow K_1$, which is bijective, and $H_0 = i(A - A^\dagger) : H \rightarrow H$. The current's quadratic form is

$$2 \operatorname{Re} \langle x, Cy \rangle + \langle y, H_0 y \rangle.$$

Make the following full invertible coordinate map from $H \oplus H$ into $K_1 \oplus H$:

$$x = (C^\dagger)^{-1}(u - \tfrac{1}{2}H_0 v), \quad y = v. \quad (\text{WD40})$$

Its form value is exactly $2 \operatorname{Re} \langle u, v \rangle$, since H_0 is Hermitian. Finally set $u = (c_+ + c_-)/\sqrt{2}$, $v = (c_+ - c_-)/\sqrt{2}$. The value becomes $\|c_+\|^2 - \|c_-\|^2$. Thus there are r positive directions, r negative directions, and exactly the $q - 2r$ radical directions in K_0 . Combined with the low-degree case, this proves

$$\boxed{\operatorname{In} i[D(M) - D(M)^\dagger] = (s, s, q - 2s), \quad s = \min(\deg D, \deg O).} \quad (\text{WD41})$$

WD34-40 retain the common factors of D and O and all nilpotent primary data. They also specify a complete sign-coordinate map for the high-degree case rather than merely a count.

7. Native substitution and the unchanged observation

The native definitions are the original RC1-2 and OCP1-7:

$$k \geq 17, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \quad 0 < \delta < 1/2, \quad \gamma > 2,$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad d\mu_k(y) = w_h^{*k}(y) dy, \quad (\text{WD42})$$

$$w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}.$$

The original source is even and positive, with the retained positive polynomial Grams. Since k is odd, the exact real factorization is

$$Q_k(y) = \prod_{b=0}^k \prod_{a=0}^{(k-1)/2} ((y - (2b-k)\gamma)^2 + (k-2a)^2\delta^2) > 0 \quad (y \in \mathbb{R}). \quad (\text{WD43})$$

This pairs a with $k-a$ inside the actual product. Every squared imaginary displacement is strictly positive, so $Q_k(0) > 0$ as well. Thus no primary is removed when defining M^{-1} . The degree q is even. At every original cutoff $q-1 \leq N \leq 2q$, both actual words

$$r_+(y) = y^{q/2}, \quad r_-(y) = y^{-q/2}$$

have full inertia $(q/2, q/2, 0)$. The finite proofs in fact hold at every $N \geq q-1$ where the source Grams are defined. They do not alter the programme's original four cutoff indices. Repeating the conjugate factors in WD43 preserves positivity and gives the same general theorems with the resulting full degree and primary multiplicities.

For the original onto observation $\Lambda : E \rightarrow B$, let

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q_B, \quad J_B = L Q_B^{-1/2}. \quad (\text{WD44})$$

Direct multiplication gives $\Lambda L = I_B$ and $J_B^* G J_B = I_B$. Thus J_B is an isometry onto $(\ker \Lambda)^\perp$. For a word factor $W = H Z H^\dagger$, the measured current is precisely

$$J_B^\dagger W J_B = Z_B H Z_B^*, \quad Z_B = Q_B^{1/2} \Lambda Z. \quad (\text{WD45})$$

This follows from $J_B^*G = Q_B^{1/2}\Lambda$, so it retains the actual section and metric. If the full inertia is $(d, d, q - 2d)$ and $m = \dim \ker \Lambda$, a positive d -dimensional subspace intersects $\text{ran } J_B$ in dimension at least $d - m$, by the dimension formula. Its restriction remains positive. No positive subspace of the restriction can exceed the maximum d in the full space. The identical argument for negative subspaces proves

$$(d - m)_+ \leq n_\pm(J_B^\dagger W J_B) \leq d. \quad (\text{WD46})$$

For $m = 8k - 16$, the native half-degree words therefore have at least

$$\frac{q}{2} - m = \frac{(k - 3)(k - 11)}{2} > 0 \quad (\text{WD47})$$

observed directions of each sign. For a rational word use $Z = Z_r$, $H = \mathcal{B} \otimes J_1$; the same proof applies with its full denominator map.

For every invertible coefficient change $S : E' \rightarrow E$, the exact transport is

$$\begin{aligned} G' &= S^*GS, & M' &= S^{-1}MS, & \Lambda' &= \Lambda S, & T' &= S^{-1}TS, & W' &= S^{-1}WS, \\ \mathfrak{W}' &= S^*\mathfrak{W}S. \end{aligned} \quad (\text{WD48})$$

Substitution into the definition of the metric adjoint verifies these formulas. Original conjugation transports to $S^{-1}CS$. Hence the original physical-unit map, its inverse, and its complete metric can be carried through every theorem without changing any inertia. This calculation makes no assertion about the sign of a separately prescribed observed terminal eigenclass.

8. Exact boundaries of the statements

The real-coefficient condition is essential. For $p = u + iv$, with u, v real, direct adjunction gives

$$W_p = W_u - [v(M) + v(M)^\dagger]. \quad (\text{WD49})$$

For a concrete complete example, take $Q = y^2 + 1$, the standard Gaussian probability measure, and $N = 1$. In the coefficient basis $1, y$, $G = I_2$, $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and W_y has eigenvalues $2, -2$. For the nonreal factor $D = y - i$, WD49 yields $W_D = W_y + 2I_2$, with eigenvalues $4, 0$. Thus the real-divisor theorem cannot be assigned to a nonreal pivot factor by its degree.

Positivity of Q is also doing exact work. For the same Gaussian source, $N = 1$, and $Q = y^2 - 1$, the original companion matrix is $M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so $W_y = 0$. The half-degree signature would fail, and WD11 fails precisely because Q changes sign. Finally, degree larger than $q/2$ has no unrestricted signature law: the real polynomial $p = Q$ has zero quotient action and zero current. The complete divisor formula WD41 determines that case and all other real divisors without extending WD14 beyond its proved domain.

9. Reading and result locators

The incoming sections 0, 1, and 2 were read in full, including all statements WS1–WS24 and WS21a–e. Incoming Section 1.1 is proved in WD5–14; Section 1.2 is proved in WD17–20 with the explicit extension of every nonmaximal partial unitary graph; Section 1.4 is proved in WD21–33; and Section 2 is proved in WD34–41 with its full source-space equality and explicit congruence. The native definitions were read directly from the retained RC1–2 and OCP source opening through OCP10. The existing machine-readable source-reading ledger for result028 supplied the programme routing before

those provider reads. No exhaustive literature reading is asserted. The new identities above are derived from their stated finite inner products, polynomial division, and matrix multiplication; no unread external theorem is invoked.

The companion `check_direct_word_grams.py` constructs six independent auxiliary instances from the full source moment matrix H_N , using source mass 3 and variance 2. It computes the actual quotient metric as $(J_N H_N^{-1} J_N^*)^{-1}$, where J_N here denotes the coefficient matrix of the unchanged remainder map. It does not obtain that metric from the orthogonal-polynomial boundary formula. It checks exact Hermitian rank, original conjugation, denominator congruence, coefficient ordering across G , the high-rank determinant ratio, and the resultant with repeated primaries. The examples include a repeated real pole, a high-degree divisor sharing a primary with its complement, the first cutoff, and inverse powers. Ordinary and optimized Python produce byte-identical `DIRECT_WORD_GRAM_CHECKS.json`. These auxiliary matrix calculations supplement the written proofs and are not native-period evaluations.

Whole-space polynomial signs, exact source mixing, and original arithmetic evolution

22 September 2026. Continuation on the original scalar-source quotient and its actual observation. GitHub input cut: fa55542372f01ba3a35ea20fa6eda1030404688f; terminal-response mathematical cut: 40a766ed3a8e0eadd9947f591803255b6eecf184; current/effective-cutoff proof: 5c69161ca70ee187f41df0bfe786e8b6eebed422.

This note derives finite whole-space sign results, rather than inferring the current of a marked eigenclass from the determinant of a positive energy. All adjoints, source minima, observation lifts, and zero factors are specified. It also identifies an incorrect equality in RD15 of the input: its first coefficient formula is retained and its identification with twice the kernel-volume constant is corrected in Section 8.

0. Input statements and scope

The literal scalar source is a positive measure μ on the real line with all used moments finite and with positive definite polynomial Grams. The relation Q is real monic of degree q and $Q(y) > 0$ for μ -almost every y . The original simple-quartet polynomial has this property: it is the product of its complete conjugate pairs, none on the real line. Repeated conjugate primary factors have the same property; simplicity is not used in Sections 1-6.

Set $E = C[y]/Q$, $M[p] = [yp]$, and let $J_N: P_N \rightarrow E$ be the original polynomial remainder map, for $N \geq q-1$. Its attained metric is G_N , with source-minimum section $R_N = J_N^\dagger$. Thus

$$J_N R_N = I_E, \quad R_N^\dagger R_N = I_E, \quad R_N J_N = P_{(Q\mathcal{P}_{N-q})^\perp} \quad \text{on } \mathcal{P}_N. \quad (\text{WS1})$$

For $N = q-1$ the relation subspace is zero. This dictionary retains the full original source mass. A coefficient change $S: E' \rightarrow E$ transports G to S^*GS , M to $S^{-1}MS$, Λ to ΛS , and every class to its inverse image. Each form below then changes by that displayed congruence. In particular the physical arithmetic unit may be inserted by its actual multiplication map, with its entire inverse and Gram, rather than deleted.

Let p_j denote real monic orthogonal polynomials for μ and $\omega_j = \|p_j\|_{\mu}^2$, including $\omega_0 = \mu(R)$. Put $\phi_j = p_j / \sqrt{\omega_j}$. The orthogonal polynomial recurrence is the full real recurrence, with its positive norm ratios; no probability normalization is imposed. Orthogonality gives the usual polynomial bandwidth directly: $\langle \phi_i, y^d \phi_j \rangle = 0$ for $i > j+d$ or $j > i+d$. DLMF 18.2 records the recurrence and positive-moment conventions; the specific quotient calculations below are derived explicitly.

For a real polynomial p define the Hermitian current endomorphism and form

$$W_p = i[p(M) - p(M)^\dagger G], \quad \mathfrak{W}_p = GW_p = i[Gp(M) - p(M)^*G]. \quad (\text{WS2})$$

Their inertia is the inertia of $G^{(1/2)} W_p G^{(-1/2)}$, equivalently of the coefficient form $\mathfrak{W}_p$. For the original physical action $\mathscr{A} = kI/2 + iM$, exactly

$$\mathscr{A}^*G + G\mathscr{A} - kG = \mathfrak{W}_y. \quad (\text{WS3})$$

For higher real polynomial tests no new weight is assigned to $p(M)$. They are related to the original linear current by the exact noncommutative telescoping identity

$$W_p = \sum_{j \geq 1} a_j \sum_{h=0}^{j-1} (M^\dagger)^h W_y M^{j-1-h}, \quad p(y) = \sum_j a_j y^j. \quad (\text{WS4})$$

Subtract consecutive powers to verify it; no factors of M and its adjoint are interchanged.

The quantitative native applications use the input RC22-27/OCP5-7 statements, on their written fixed-packet and fixed-period domains:

$$\begin{aligned} M &= C + \epsilon f e^\dagger, \quad C = C^\dagger, \quad e \perp f, \quad \|e\| = \|f\| = 1, \\ \|C\| &\leq 2(N+1)\sqrt{u_k/\ell_k}, \end{aligned} \tag{WS5}$$

$$\begin{aligned} \log \epsilon &= q\psi(t_N) + O_{h,\varpi}(k \log(q+2)), \quad t_N = (N+1-q)/q, \\ \log(u_k/\ell_k) &= O_h(k + \log(q+2)). \end{aligned} \tag{WS6}$$

The last ratio is applied at every actual extra degree used below; the source FW7 states the bound for arbitrary finite degree. The input does not need a new source cutoff to represent an outgoing class: its entire degree- N minimum is used.

The observed two-column estimates already established in OCP and RC are

$$\begin{aligned} U_B &= [Q_B^{1/2} \Lambda e, Q_B^{1/2} \Lambda f], \quad 0 \prec U_B^* U_B \preceq I_2, \\ \det(U_B^* U_B) &\geq \Theta_N > 0, \end{aligned} \tag{WS7}$$

$$\begin{aligned} -\log \Theta_N &= O(k \log q), \quad \alpha_N := \|(I - L\Lambda)e\|_G^2 \\ &\leq e^{-2q\psi(t_N) + O(k \log q)}. \end{aligned} \tag{WS8}$$

The actual conductor margin in OCP19-23 includes its complex off-diagonal determinant product. Nothing in this note substitutes individual column norm bounds for WS7. The source's terminal-class norm guard is stronger and is not needed for the whole-space results.

1. Every real polynomial word up to half the packet degree

Let p be any real polynomial of degree d , $1 \leq d \leq q/2$, with leading coefficient $a_d \neq 0$. It may have arbitrary lower coefficients. Define the incoming and outgoing maps, with their original source scalars,

$$I_d = [J_N \phi_{N-d+1}, \dots, J_N \phi_N], \quad O_d = [[\phi_{N+1}], \dots, [\phi_{N+d}]], \tag{WS9}$$

where every outgoing class is evaluated in G_N , not in G_{N+d} . Define the real d -by- d matrix

$$B_p[r, s] = \langle \phi_{N+r}, p(y) \phi_{N-d+s} \rangle_\mu, \quad 1 \leq r, s \leq d. \tag{WS10}$$

Degree proves $B_p[r, s] = 0$ for $r > s$. Its complete diagonal is

$$B_p[s, s] = a_d \sqrt{\omega_{N+s}/\omega_{N-d+s}} \neq 0. \tag{WS11}$$

Thus B_p is invertible.

For an actual class x , multiply its complete minimum representative $R_N x$ by p . Its part in P_N compresses to the selfadjoint operator $C_p = R_N^\dagger P_N p(y) R_N$. The outgoing part uses only the last d source coordinates, and the exact adjoint identity in WS1 gives

$$p(M) = C_p + O_d B_p J_d^{\dagger G}, \quad C_p = C_p^\dagger. \tag{WS12}$$

Set

$$Z_p = [I_d, O_d B_p], \quad J_d = \begin{pmatrix} 0 & -iI_d \\ iI_d & 0 \end{pmatrix}.$$

Then

$$\boxed{W_p = Z_p J_d Z_p^{\dagger G}}. \tag{WS13}$$

This contains every source multiplication cross term of WS4 in one exact boundary matrix.

1.1 The boundary map is injective, including repeated primaries

Consider a source polynomial F in the span of the $2d$ consecutive ϕ 's in WS9 whose class is zero. It equals Qh , with $\deg h \leq N+d-q \leq N-d$. On the other hand F is orthogonal to $P_{-(N-d)}$. Therefore

$$0 = \langle h, F \rangle_\mu = \int Q(y) |h(y)|^2 d\mu(y). \quad (\text{WS14})$$

Positivity of Q on the measure and positive polynomial Grams imply $h=0$. Independence of the source ϕ 's gives $F=0$. Hence $[I_d, 0_d]$, and therefore Z_p , has full column rank $2d$. This proof uses no separation or simplicity of the roots of Q .

Put $R_p = Z_p^* G Z_p > 0$ and $V_p = Z_p R_p^{-1/2}$. Then $V_p^\dagger V_p = I_{(2d)}$, and on its image

$$V_p^\dagger W_p V_p = R_p^{1/2} J_d R_p^{1/2}.$$

The orthogonal complement is the full radical. Congruence with the invertible $R_p^{1/2}$ proves

$$\boxed{\text{In}(W_p) = (d, d, q - 2d).} \quad (\text{WS15})$$

In particular the actual native q is even, so the explicitly specified word $p(y) = y^{(q/2)}$ has inertia $(q/2, q/2, 0)$ at every original canonical cutoff. It has no hidden zero-current direction. This higher word is related to the original linear action by WS4; it is not identified with that linear action. These are numbers of positive, negative and zero eigenvalues on the complete original E . They do not depend on period orientation, on the lower coefficients of p , or on cutoff N within the stated domain.

1.2 All vectors and all maximal isotropic subspaces

For x in E set $t = Z_p^\dagger x = (a, b)$, and

$$c_+ = (a - ib)/\sqrt{2}, \quad c_- = (a + ib)/\sqrt{2}.$$

Every t occurs: the exact right inverse is $x = Z_p R_p^{-1} t + x_0$. Every x has the unique decomposition

$$x = Z_p R_p^{-1} t + x_0, \quad x_0 \in \ker Z_p^\dagger,$$

with squared norm $t^* R_p^{-1} t + \|x_0\|_G^2$. Direct multiplication proves the entire sign formula

$$\boxed{\langle x, W_p x \rangle_G = 2\Im(a^* b) = \|c_+\|^2 - \|c_-\|^2.} \quad (\text{WS16})$$

Thus the positive, negative and null sets are exactly the corresponding norm comparisons in these specified coordinates. The radical is $a=b=0$; an isotropic vector need not lie there.

Let $S = 2^{-1/2} [[I, I], [iI, -iI]]$, so $t = S(c_+, c_-)$ and $S^* J_d S = \text{diag}(I, -I)$. Every maximal totally isotropic complex subspace is exactly

$$\{Z_p R_p^{-1} S(h, Uh) + x_0 : h \in \mathbb{C}^d, x_0 \in \ker Z_p^\dagger\}, \quad U^* U = I_d. \quad (\text{WS17})$$

It has dimension $q-d$. To prove completeness, a maximal isotropic space contains the radical. In the remaining signature- (d, d) space its projection onto the positive coordinates is injective, so its dimension is at most d ; at dimension d it is the graph of a norm-preserving, hence unitary, map U . WS17 restores the exact source metric and inverse coordinate map.

Since Q , μ and p are real, coefficient conjugation preserves G and sends W_p to $-W_p$. Thus its nonzero eigenvalues are paired exactly as $+\sigma_j, -\sigma_j$. Conjugation is transported by the actual physical-unit map under a coefficient change; it is not presumed to be entrywise conjugation in an unrelated physical frame.

1.3 Observations and all their zero cases

For any given onto Lambda use its actual minimum L, metric Q_B, and isometry J_B. Then

$$W_{p,B} = J_B^\dagger W_p J_B = Z_B J_d Z_B^*, \quad Z_B = Q_B^{1/2} \Lambda Z_p. \quad (\text{WS18})$$

Let C_B = Z_B^* Z_B, including its full nullspace. On its positive range V = Z_B C_B^{\{+1/2\}} is an isometry, and

$$W_{p,B} = V [C_B^{1/2} J_d C_B^{1/2}] V^*. \quad (\text{WS19})$$

Its other directions are zero. This is a complete matrix of order at most 2d for the whole observed sign problem, with no replacement of a zero block by an inverse. The original observed vector b has form value (Z_B^* Q_B^{(1/2)} b)^* J_d (Z_B^* Q_B^{(1/2)} b).

Restriction min-max gives the explicit count bounds

$$(d - m)_+ \leq n_\pm(W_{p,B}) \leq d, \quad m = \dim \ker \Lambda. \quad (\text{WS20})$$

In particular, at every native degree $m < d \leq q/2$, both observed signs are present for every original observation of that rank. At the native half-degree word, the lower bound is exactly $q/2 - m = (k-3)(k-11)/2$, positive for every admitted $k \geq 17$. Thus its original observed current has at least that many directions of each sign, independently of the period orientation. For $d=1$ the stronger exact $(1, 1, \dim B - 2)$ count is already the source OCP theorem on its full conductor-area domain. It is not claimed again as a new two-column estimate.

The observed word in WS18 is the measured full word $\Lambda p(M)L$. The exact difference from applying p to the intrinsic compressed generator is retained. For $d \geq 2$, with $M_B = \Lambda p(M)L$ and $Qp = I - L \Lambda p$,

$$\Lambda M^d L - M_B^d = \sum_{j=0}^{d-2} M_B^j \Lambda M Qp M^{d-1-j} L. \quad (\text{WS20a})$$

Insert $I = L \Lambda p + Qp$ after the first M and iterate to prove this identity. Each right-hand factor retains all subsequent hidden excursions. Thus no nonlinear observation/action interchange is used in the sign calculation.

For a complex polynomial $p = u + iv$, u, v real, the exact relation is

$$W_p = W_u - [v(M) + v(M)^\dagger]. \quad (\text{WS21})$$

The second term is retained. WS15 is a real-polynomial theorem; WS21 is the precise additional term for a nonreal pivot word.

1.4 Real rational words: full sign classification through denominator maps

Let $r(y) = p(y)/s(y)$ have real coefficients, with $\gcd(s, q) = 1$. Cancel the actual polynomial gcd of p, s first. Denote the reduced numerator and denominator again by p, s and put $d = \max(\deg p, \deg s)$. For $1 \leq d \leq q/2$, the entire rational action

$$T_r = p(M)s(M)^{-1}$$

is well-defined on the original quotient, with all primary multiplicities. The denominator may have real or repeated poles; no rational function is integrated against the source measure in this proof. Only polynomial source inner products and the invertible quotient multiplication map $s(M)$ are used.

Define the real symmetric Bezout coefficient matrix B by the exact polynomial identity

$$\frac{s(x)p(y) - p(x)s(y)}{y - x} = \sum_{i,j=0}^{d-1} B_{ij} x^i y^j. \quad (\text{WS21a})$$

It is invertible for the reduced pair. A proof retaining repeated roots is as follows. Arrange, by swapping p, s and reversing the sign if necessary, that $\deg s = d$. In the quotient algebra $C[x]/s$, multiplication by p is invertible by the Euclidean polynomial identity. Put $c_s(x, y) = (s(y) - s(x))/(y - x)$. Modulo $s(x)$, WS21a is exactly $-p(x)c_s(x, y)$: multiply the proposed identity by the monic polynomial $y - x$ in the ring $(C[x]/s)[y]$, where that multiplication is injective by its leading coefficient one. The coefficient matrix of c_s is anti-triangular with the nonzero leading coefficient of s on its anti-diagonal. Thus B is the product of an invertible quotient multiplication matrix and this invertible coefficient matrix. No simple-root interpolation is used.

Choose the exact rank-two linear-current factor $W_y = Z_1 J_1 Z_1^\dagger$ from WS13. Define

$$\mathcal{Z}_d = [Z_1, M^\dagger Z_1, \dots, (M^\dagger)^{d-1} Z_1].$$

WS4 for y^d gives

$$W_{y^d} = \mathcal{Z}_d (R_d \otimes J_1) \mathcal{Z}_d^\dagger,$$

where R_d is the d -by- d anti-diagonal permutation matrix. WS15 already proves rank $2d$ for the left side; consequently $\mathcal{Z}_d$ has rank $2d$. This proves the needed complete Krylov independence without a new genericity assumption.

Multiply the actual rational current form by its denominator on both sides. The exact identity is

$$\begin{aligned} s(M)^* \mathfrak{W}_r s(M) &= i[s(M)^* G p(M) - p(M)^* G s(M)] \\ &= G \mathcal{Z}_d (B \otimes J_1) \mathcal{Z}_d^{\dagger G}. \end{aligned} \tag{WS21b}$$

To verify the second line, multiply WS21a by $y - x$ and substitute each monomial $x^i y^j$ by $(M^*)^i G M^j$. Left and right multiplication are the explicitly commuting operations on the middle coefficient matrix; M and G themselves are not assumed to commute. The middle difference $y - x$ supplies exactly $i(GM - M^*G)$.

The denominator congruence is invertible. Equivalently, with

$$Z_r = (s(M)^{\dagger G})^{-1} \mathcal{Z}_d, \quad W_r = Z_r (B \otimes J_1) Z_r^\dagger, \tag{WS21c}$$

the column map Z_r has rank $2d$. Every nonzero real eigenvalue of B contributes one positive and one negative eigenvalue when tensored with J_1 . Therefore

$$\boxed{\text{In } i(T_r - T_r^\dagger) = (d, d, q - 2d).} \tag{WS21d}$$

A real constant rational word has the zero current. Every nonconstant real rational word of the stated reduced degree has this complete whole-space signature. Repeated poles, collisions among denominator poles, and arbitrary real lower coefficients do not change it. An actual common numerator/denominator factor lowers the reduced degree by its full degree; it contributes the corresponding additional zero directions. Invertibility of $s(M)$ is never retained across a common factor with Q by deleting that primary.

This includes the programme's original inverse powers: $r(y) = y^{-d}$ is defined because the original $Q_k(0)$ is nonzero, and its current has exactly d signs of each kind for $d \leq q/2$. The exact connection is $W_-(M^{(-d)}) = -M^{(-d \dagger)} W_-(M^d) M^{(-d)}$. At $d = q/2$, every one of the q directions is nondegenerate for both words.

All sign witnesses and null spaces can be recovered from WS21c by its full positive Gram and a real orthogonal diagonalization of B , followed by the fixed two-by-two eigen-isometry of J_1 . For each negative B eigenvalue interchange its two signs. This constructs the same hyperbolic coordinates as WS16–17 through specified invertible matrices, with the full rational denominator retained.

At the full-rank endpoint $q = 2d$ there is also an exact determinant identity:

$$\boxed{\det W_r = \frac{(\det B)^2}{|\det s(M)|^2} \det W_{y^d}.} \tag{WS21e}$$

Indeed $\mathcal{Z}_d$ is square invertible; its complete metric Gram cancels between the two determinant factorizations, while the denominator contributes $|\det s(M)|^{(-2)}$. The factor $\det s(M) = \text{Res}(Q, s)$ retains every original primary multiplicity; the equality follows by triangularizing the quotient multiplication on its full primary blocks, whose diagonal values are s at those roots. Thus for a fixed rational word at this rank, its full four-cutoff logarithmic determinant return equals that for M^d exactly. No uncomputed asymptotic coefficient for that high-degree return is assigned by the equality.

2. Real divisor words of every degree

Factor the same real positive relation as $Q=D \cdot O$, with real monic D, O . Put $d_D = \deg D$, $r = \deg O$, so $q = d_D + r$. No coprimality is required. For the actual word $T=D(M)$, exact polynomial cancellation gives

$$\begin{aligned} \ker T &= \mathcal{OP}_{d_D-1}, & \text{im } T &= D\mathcal{P}_{r-1}, \\ \text{rank } T &= r. \end{aligned} \tag{WS22}$$

When $d_D \leq r$, WS15 applies directly. Consider $d_D \geq r$. Suppose $[D \ b]$ is orthogonal to $\ker T$, and let $F=D \cdot b + Qh=Dg$ be its entire degree- N minimum. It is orthogonal to the full relation space. Together with the assumed kernel orthogonality, this makes F orthogonal to

$$\mathcal{OP}_{d_D-1} + Q\mathcal{P}_{N-q} = \mathcal{OP}_{N-r}.$$

The equality follows by division by the monic D . Here $\deg g \leq N - d_D \leq N - r$, so Og is one of those actual source polynomials. Therefore

$$0 = \langle Og, Dg \rangle_\mu = \int Q|g|^2 d\mu,$$

forcing $g=0$. Thus

$$\text{im } T \cap (\ker T)^{\perp_G} = 0. \tag{WS23}$$

In the exact orthogonal splitting $\ker T$ plus its complement, write $T = \begin{bmatrix} [0, B], [0, A] \end{bmatrix}$. The complement has dimension r . WS23 proves that B is injective. On $\text{ran } B$ plus $(\ker T)^{\perp_G}$, the current has the complete form

$$\begin{pmatrix} 0 & iB \\ -iB^\dagger & i(A - A^\dagger) \end{pmatrix}.$$

The off-diagonal block has rank r . A triangular congruence removes the last diagonal block and then an invertible change by B gives r hyperbolic pairs; all $q - 2r$ orthogonal kernel directions are zero. Hence the complete result, covering both degree orders, is

$$\boxed{\text{In } i(D(M) - D(M)^\dagger) = (s, s, q - 2s), \quad s = \min(\deg D, \deg O).} \tag{WS24}$$

A degree-zero real factor gives the zero current, as does $D=Q$; these are the $s=0$ cases of WS24. The formula includes shared primary factors and repeated conjugate roots. For a complex pivot factor use WS21; it is not made real by changing its label. For example $Q=y^2+1$, the Gaussian source on P_1 with $G=I$, and $D=y-i$ give one positive current and one zero. The explicit difference is $W_D=W_y+2I$, verifying why the real-factor domain matters.

3. The complete boundary Gram and the evaluated signed volume

Continue with the real degree- d word, $2d \leq q$. Let U_j be the real monic orthogonal polynomial for the full measure $Q^2 d\mu$, and $nu_j = ||QU_j||_\mu^2$. Its normalized

relation columns are $r_n = Q U_{(n-q)}/\sqrt{\nu_{(n-q)}}$, for $n \geq q$. In the high source rows $\phi_{(N-d+1)}, \dots, \phi_{(N+d)}$, the only contributing relation columns are the last at most d old ones and the d new ones. Their entire coefficient matrix is

$$\begin{aligned} R_H &= \begin{pmatrix} X & Y \\ 0 & Z_0 \end{pmatrix}, \\ (Z_0)_{rs} &= \langle \phi_{N+r}, r_{N+s} \rangle_\mu, \\ (Z_0)_{ss} &= \sqrt{\omega_{N+s}/\nu_{N+s-q}}. \end{aligned} \tag{WS25}$$

Z_0 is upper triangular and invertible. X can have fewer than d columns at the first cutoffs; the formula uses its actual width.

The high-source Gram at the larger minimum $G_{(N+d)}$ is $I - R_H R_H^*$, by projection against all orthonormal relation columns. The complete covariance update is $G_{(N+d)}^{(-1)} = G_N^{(-1)} + 0_d 0_d^*$. Apply the matrix inversion identity to this full update. If V denotes $I_d^* G_N 0_d$ and D denotes $0_d^* G_N 0_d$, the new lower block is $D(I+D)^{(-1)} = I - Z_0 Z_0^*$; thus $(I+D)^{(-1)} = Z_0 Z_0^*$. The new cross block is $V(I+D)^{(-1)} = -Y Z_0^*$, so $V = -Y Z_0^*$. Returning the upper block gives the exact old-cutoff Gram

$$\boxed{{}^*G_N[I_d, O_d] = \begin{pmatrix} I - X X^* & -Y Z_0^{-1} \\ -Z_0^{-*} Y^* & (Z_0 Z_0^*)^{-1} - I \end{pmatrix}}. \tag{WS26}$$

All old and new source relations and all cross terms are included. The larger cutoff has been used only in this exact algebraic identity; the final metric in WS26 is G_N .

In the balanced fixed-cutoff frame

$$\begin{aligned} F &= [I_d, O_d Z_0], \quad K_0 = F^* G_N F = \\ &\begin{pmatrix} I - X X^* & -Y \\ -Y^* & I - Z_0^* Z_0 \end{pmatrix} > 0, \end{aligned}$$

put $D_p = Z_0^{(-1)} B_p$. The current becomes

$$W_p = F \begin{pmatrix} 0 & -i D_p^* \\ i D_p & 0 \end{pmatrix} F^\dagger G. \tag{WS27}$$

These formulas specify the exact balancing map and metric; F is not declared orthonormal.

If $\delta = \max(|X|, |Y|, |Z_0|)$ and $\rho = \delta + \delta^2 < 1$, the complete Gram obeys $(1 - \rho)I \leq K_0 \leq (1 + \rho)I$. Generalized min-max, on the positive and negative spectral subspaces of the numerator in WS27, gives

$$\begin{aligned} (1 - \rho)\sigma_j(D_p) &\leq \lambda_j^+(W_p), \quad -\lambda_j^-(W_p) \leq (1 + \rho)\sigma_j(D_p), \\ 1 &\leq j \leq d. \end{aligned} \tag{WS28}$$

The same factor can bound both signs because the denominator is positive. The entire nonzero determinant is even more explicit and needs no small- ρ condition:

$$\boxed{\text{pdet } W_p = (-1)^d |a_d|^{2d} \left[\prod_{s=1}^d \frac{\nu_{N+s-q}}{\omega_{N-d+s}} \right] \det K_0}. \tag{WS29}$$

Indeed the determinant of the small numerator is $(-1)^d |\det D_p|^2$; multiplication by the entire positive Gram contributes $\det K_0$. WS11 and WS25 cancel exactly the intermediate norms $\omega_{(N+s)}$. Thus the product of all positive eigenvalues, squared,

is the right side's absolute value. It depends on the lower coefficients of p only through **none of its factors**. For fixed leading coefficient, changing any real lower coefficient changes neither this full signed product nor the inertia. It can change individual eigenvalues; those are not identified by their product.

3.1 A finite native Gram guard

The finite guard in WS28 can be bounded without an unevaluated angle matrix. Use the original complete source comparison at degree $2q+d+1$, with ratio $\kappa_k = u_k / \ell_k$. The source FW7 has this degree explicitly available. Put

$$R_* = \max\{1, k\sqrt{\delta^2 + \gamma^2}, 2(N+d+1)\sqrt{\kappa_k}\}, \quad B_* = 2(N+d)R_*,$$

$$\zeta_* = \max_{\max(q, N-d+1) \leq n \leq N+d} \sqrt{\omega_n / \nu_{n-q}}, \quad (WS30)$$

$$\delta_* = d\sqrt{\kappa_k} B_*^{2d} \zeta_*.$$

This is a finite bound $\delta \leq \delta_*$ for all the actual blocks in WS25. Here is its proof. Compressed real multiplication places every root of the used p_n and U_j inside the multiplication norm bound in R_- ; *the latter measure is $Q^2 dm_u$, so its Rayleigh quotient is the ratio of the actual norms of yQP and QP . The source comparison and Gamma multiplication bound apply through their actual degrees. The roots of Q are also bounded by R_- .*

Consequently, a coefficient h positions below the leading coefficient of either monic polynomial is bounded by $(n R_-^*)^h$. In the expansion $Q U_{(n-q)} = \sum_i c_{(i,n)} p_i$, triangular elimination gives $|c_{(n-h,n)}| \leq (2n R_-^*)^h$: induct on h using the bound $1 + \sum_{j < h} 2^j = 2^h$. A normalized relation coefficient is $c_{(i,n)} \sqrt{\omega_i / \omega_n} \sqrt{\omega_n / \nu_{(n-q)}}$. The indices differ by at most $2d$; the same Gamma source comparison gives $\sqrt{\omega_i / \omega_n} \leq \sqrt{\kappa_k}$ in this high-index range. The matrix norm is at most d times its maximum entry. This proves WS30, with no omission of an off-diagonal block.

For the admitted native simple quartet, use RC22–24's full monic estimate at every actual n in WS30:

$$\log \frac{\nu_{n-q}^\sigma}{h_n} = 2q\psi((n-q)/q) + O_h(\log(q+2)),$$

$$h_n = \sqrt{2\pi} n! (1/2)_n.$$

The original arithmetic norm comparison restores ω_n and $\nu_{(n-q)}$ with logarithmic error $O_h(k + \log(q+d+2))$. The profile's endpoint derivative bound gives a shift error $O_h(d \log(q+2))$. Therefore

$$\log \delta_* \leq -q\psi(t_N) + O_h\{k \log(q+2) + d[k + \log(q+2)]\}. \quad (WS31)$$

For every integer sequence $d = o(k)$, this is negative by order q , uniformly over the original N window, because the stated profile has positive minimum there. The complete Gram K_0 therefore approaches I in this explicit balanced frame; the original G_N is not set to I .

Combining WS29, the exact Gamma factorial ratios, and the same finite source comparisons proves

$$\sum_{j=1}^d \log \sigma_j(W_p) = d \log |a_d| + dq\psi(t_N) + O_h\{dk \log(q+2) + d^2 \log(q+2)\}. \quad (WS32)$$

Here the σ_j are the d positive eigenvalues, with the negative eigenvalues exactly their opposites. The error is uniform in all real lower coefficients, even when they vary: their cancellation is the exact identity WS29. The leading coefficient may depend on k but is the same at every cutoff.

Taking the original four signs, whose parameters are exactly $0, 1/q, 1, 1+1/q$, yields

$$\mathcal{R} \sum_{j=1}^d \log \sigma_j(W_p) = 2d[\psi(0) - \psi(1)]q + O_h\{dk \log(q+2) + d^2 \log(q+2)\}. \quad (\text{WS33})$$

The endpoint error is obtained by integrating the logarithmic bound for ψ' near zero and its bounded derivative near one. No adjacent source degree is merged without its error. After division by dq the error tends to zero throughout $d=o(k)$.

For any fixed real polynomial p , WS28 and the triangular matrices give the individual laws $\log \sigma_j = q \psi(t_N) + 0_-(h, p, d)(k \log q)$. One verification bounds B_p by the full multiplication norm of $p(y)$, bounds its diagonal below with WS11, and uses the finite upper-triangular inverse recursion. The same recursion for Z_0 , whose diagonal logarithms are $-q \psi + 0(k \log q)$, bounds its scaled matrix and inverse by $\exp[0_-(h, d)(k \log q)]$. These estimates are not uniform in arbitrarily large lower coefficients; WS32–33 are uniform because of their exact determinant cancellation.

The same product evaluates an unavoidable whole-space correction cost. For a nonnegative scalar correction to the Hermitian current, the least value is $b_{\text{opt}} = \max_j \sigma_j(W_p)$. WS29 gives the finite bound

$$b_{\text{opt}} \geq |a_d| \left[\det K_0 \prod_{s=1}^d \frac{\nu_{N+s-q}}{\omega_{N-d+s}} \right]^{1/(2d)}.$$

Thus every monic real word with $d=o(k)$, without any lower-coefficient restriction, has $\log b_{\text{opt}} \geq q \psi(t_N) - 0_-(k \log q + d \log q)$ in the native family. A fixed polynomial scalar allowance cannot dominate this canonical whole-space form. For a general positive operator correction Z , the least possible trace is $\sum_j \sigma_j$, attained by the negative spectral part of W_p ; the least possible rank is d . Indeed restrict $W_p + Z \geq 0$ to its entire negative spectral subspace and take its trace, and then use rank-nullity on that same subspace. These are correction costs for this specified Hermitian form in G_N , not an assertion that an arbitrary source enlargement realizes the minimizing correction.

3.2 The full physical Hermitian part and all exterior powers

For $p=y$, the full current is exactly ϵJ_1 on its two-dimensional active plane. Thus its nonzero eigenvalues are $+\epsilon$ and $-\epsilon$. The full uncentered physical form is $G(kI + W_y)$ and has eigenvalues $k+\epsilon, k-\epsilon, k$ in the actual metric. On the explicit finite guard $\epsilon > k$ its inertia is $(q-1, 1, 0)$. The native estimates WS6 make this guard hold eventually, uniformly over the source window. The observed form has the same count $(\dim B - 1, 1, 0)$ on the already-proved guard $\epsilon \geq \Theta_N$. These are values of the displayed uncentered forms, not the sign of a separately marked vector.

Let $\mathscr{A}^{[r]}$ be the actual derivation action on the r -th exterior power, not the multiplicative matrix $\wedge^r \mathscr{A}$. The induced metric is $\wedge^r G$, and

$$\begin{aligned} (\mathscr{A}^{[r]})^* \wedge^r G + (\wedge^r G) \mathscr{A}^{[r]} - rk \wedge^r G \\ = (\wedge^r G) W_y^{[r]}. \end{aligned} \quad (\text{WS34})$$

The exact original orthonormal eigen-isometry of W_y induces its exterior isometry. Selecting $+\epsilon$ without $-\epsilon$ gives $\text{binom}(q-2, r-1)$ positive eigenvalues $+\epsilon$; selecting the negative alone gives the same number $-\epsilon$. Selecting both or neither gives zero. Hence

$$\boxed{\text{In } W_y^{[r]} = \left(\binom{q-2}{r-1}, \binom{q-2}{r-1}, \binom{q-2}{r} + \binom{q-2}{r-2} \right)}. \quad (\text{WS35})$$

Use binomial zero outside its usual range. The uncentered form is $\text{rk}I + W_y^{[r]}$; on $\epsilon > \text{rk}$ its negative count is $\text{binom}(q-2, r-1)$ and every other direction is positive. This guard holds eventually simultaneously for $1 \leq r \leq q$, since ϵ grows exponentially and $\text{rk} \leq qk$. At $r=q$ the centered current is zero and the uncentered form is positive.

3.3 Every block of the original linear Hermitian current

In the exact observed/kernel isometries let $U_B = [P_e, P_f]$ and $U_K = [Q_p \ e, Q_p \ f]$, retaining their actual codomains. Their Grams are

$$C_B = \begin{pmatrix} a & r \\ \bar{r} & b \end{pmatrix}, \quad C_K = I_2 - C_B.$$

The observed eigenvalues are the existing OCP values $\epsilon [\Im r \pm \sqrt{(1-a)(1-b) - (\Re r)^2}]$. On the entire original kernel the complete list, before zero padding, is

$$\epsilon \left[\Im r \pm \sqrt{(1-a)(1-b) - (\Re r)^2} \right]. \quad (\text{WS35a})$$

It follows by multiplying the full matrices $J_1 \ C_K$, including the complex off-diagonal entries. At $\det C_K > 0$ there is one sign of each kind; at determinant zero the possible remaining eigenvalue is $2\epsilon \Im r$, including its actual zero case. When $C_K = 0$, the whole hidden current vanishes.

The mixed **Hermitian** current is the off-diagonal block operator built from $H = \epsilon U_B J_1 U_K^\dagger$. It has eigenvalues $+\sigma_j(H), -\sigma_j(H)$ and the full remaining zeros. Their squares are the nonzero eigenvalues of

$$\epsilon^2 C_B^{1/2} J_1 C_K J_1 C_B^{1/2} \succeq 0. \quad (\text{WS35b})$$

On the established native area domain $C_B > 0$, its rank is $\text{rank } C_K$. Thus the mixed Hermitian current has inertia $(\text{rank } C_K, \text{rank } C_K, q - 2\text{rank } C_K)$. These are exact classifications of all three blocks; they do not add their signatures to obtain the signature of their sum, since the retained mixed blocks couple the two subspaces.

4. Complete whole-space signs of the mixed quartic

Keep the original orthogonal projection $P = L \Lambda$ and $Q_p = I - P$. In the actual isometries of B and K define the two full arithmetic mixed blocks

$$C = Q_p M|_{K^\perp} : K^\perp \rightarrow K, \quad D = P M|_K : K \rightarrow K^\perp. \quad (\text{WS36})$$

For $x = k + b$ with k in K and b in $K^\perp$, the physical cross pairings are

$$\begin{aligned} z(x) &= \langle k, \mathcal{A}b \rangle_G = i \langle k, Cb \rangle_G, \\ w(x) &= \langle b, \mathcal{A}k \rangle_G = i \langle b, Dk \rangle_G. \end{aligned}$$

The scalar centre cancels by orthogonality. Define the actual quartic $\chi(x) = \text{Re}(\text{conj}(z(x))w(x))$.

The exact G-unitary map $U_\theta = P + \exp(i\theta)Q$ changes the pair as

$$\begin{aligned} z(U_\theta x) &= e^{-i\theta} z(x), & w(U_\theta x) &= e^{i\theta} w(x), \\ \chi(U_\theta x) &= \Re(e^{2i\theta} \overline{z(x)} w(x)). \end{aligned} \tag{WS37}$$

It also satisfies $\Lambda U_\theta = \Lambda$: the observed value, both component norms, and both pure observed/hidden currents are preserved. Every nonzero product orbit therefore attains both signs, with extrema $+|zw|$ and $-|zw|$ and with exact intervening zeros.

The complete degeneracy classification is: χ is identically zero exactly when $C=0$ or $D=0$. One direction is immediate. For the other, C and $D^\dagger$ are nonzero linear maps on $K^\perp$, so their two proper kernels cannot cover that complex vector space. Select b outside both; then select k outside the two proper orthogonality hyperplanes to Cb and $D^\dagger b$. This constructs z and w both nonzero. All such choices can be made by a finite pencil $b_1 + t b_2$, then $k_1 + s k_2$, avoiding the at most two roots of each pair of linear functionals.

On the original bounded-observation domain, write $K_j = \text{intersection}_{(h=0)}^j \ker(\Lambda M^h)$. The source proves $\dim K_8 \leq 128$ and the decreasing layer ranks. Thus

$$\text{rank } D = \text{rank}(\Lambda M I_K) \geq (m - 128)/8 = k - 18.$$

Also $C - D^\dagger = Q(P - M^\dagger)P$ has rank at most two by WS5. Therefore for every admitted $k \geq 21$, both C and D are nonzero. WS37 proves that the actual whole-space mixed quartic takes positive, negative and zero values at every cutoff on that domain. This is not a proposed sign for a fixed eigenclass.

Its failure to preserve that eigenclass is an explicit arithmetic defect:

$$MU_\theta - U_\theta M = (e^{i\theta} - 1)(PMQ - QPMP). \tag{WS38}$$

Thus rotating k cannot be used to change a prescribed terminal eigenclass while claiming it is the same input.

For a complete Hermitian realization of the quartic, let $S = \text{Pi_Sym2}(B) - (C^\dagger \text{ tensor } D) - \text{Pi_Sym2}(K)$. With the actual induced symmetric-square metrics,

$$\chi(k + b) = \Re\langle b \otimes b, S(k \otimes k) \rangle.$$

The block Hermitian operator on $\text{Sym}^2 K$ plus $(K \text{ tensor } B)$ plus $\text{Sym}^2 B$, with only off-diagonal blocks $S/2, S^\dagger/2$, has inertia $(\text{rank } S, \text{rank } S, \dim \text{Sym}^2 B - 2\text{rank } S)$. Its restriction to the actual pure-square image is exactly the original χ ; WS37 proves both signs on that image itself, rather than inferring them from arbitrary symmetric tensors.

5. Original source activation has an affine sign pencil

Let the actual covariance, including its original complete normal-source correction, be

$$C_\alpha = G_\alpha^{-1} = (1 - \alpha)C_0 + \alpha C_1 > 0, \quad 0 \leq \alpha \leq 1.$$

Fix the original action word T and the original onto Λ . Define the fixed-coordinate dual current

$$S_\alpha = i(TC_\alpha - C_\alpha T^*) = (1 - \alpha)S_0 + \alpha S_1. \tag{WS39}$$

The exact full and observed congruences are

$$\boxed{\begin{aligned} C_\alpha i(G_\alpha T - T^* G_\alpha) C_\alpha &= S_\alpha, \\ L_\alpha^* i(G_\alpha T - T^* G_\alpha) L_\alpha \\ &= Q_{B,\alpha} \Lambda S_\alpha \Lambda^* Q_{B,\alpha}. \end{aligned}} \quad (\text{WS40})$$

Here $L_\alpha = C_\alpha \Lambda^{-1} Q_{B,\alpha}$ and the quotient metric is formed from that same covariance. No section is frozen. The actual maps taking a dual vector to its original primal sign witness are $x = C_\alpha z$ in E and $b = Q_{B,\alpha}^{-1} z$ in B .

For any finite collection of source covariances, the same proof makes the dual current an affine Hermitian function of all weights. Thus its positive-semidefinite region is exactly the intersection of the parameter domain with that affine matrix inequality. On a covariance polytope, positivity on the entire whole space is equivalent to positivity at every vertex: a convex combination of positive forms is positive, and the converse includes each vertex. A fixed marked class has value $b^* Q_{B,\alpha} (\Lambda S_\alpha \Lambda^{-1}) Q_{B,\alpha} b$; its dual coordinate moves. This formula is the exact reason the whole-space vertex statement does not validate vertex sampling of an unrelated marked scalar.

5.1 A finite reduction that keeps a singular joint current

For a real polynomial word of degree $d \leq q/2$, the canonical dual current has the factor $S_0 = U J_d U^*$, where $U = Z_p$ in E or $U = \Lambda Z_p$ in B . The latter can be rank deficient; the following formula allows that.

For $\alpha > 0$ put $t = (1 - \alpha)/\alpha$. The inertia to compute is that of $H(t) = S_1 + t U J_d U^*$. Use an actual unitary split of the fixed coefficient space into $\text{ran } S_1$ plus $\ker S_1$:

$$\begin{aligned} S_1 &= \text{diag}(D, 0_z), \quad D = D^* \text{ invertible}, \quad U = \begin{pmatrix} U_1 \\ U_0 \end{pmatrix}, \\ C &= U_1^* D^{-1} U_1. \end{aligned}$$

Put $s = \text{rank } U_0$. Choose an isometry onto $\text{ran } U_0$, so U_0 has its s nonzero rows V and $z - s$ zero rows. Let N be an isometry from $\mathbb{C}^{(2d-s)}$ onto $\ker V$. The whole inertia is

$$\boxed{\begin{aligned} \text{In } H(t) &= \text{In } D + \text{In}[-N^*(J_d + tC)N] \\ &\quad + (s, s, z - s) - (d, d, 0). \end{aligned}} \quad (\text{WS41})$$

The additions and subtraction are componentwise; the formula as a whole has nonnegative entries and total original dimension. The reduced Hermitian pencil has order at most $2d$.

For a full proof, border $H(t)$ before eliminating anything:

$$\mathcal{B}_t = \begin{pmatrix} D & 0 & U_1 \\ 0 & 0 & U_0 \\ U_1^* & U_0^* & -J_d/t \end{pmatrix}.$$

Eliminating the invertible final block gives $\text{In } \mathcal{B}_t = \text{In } H(t) + (d, d, 0)$. Eliminating D instead gives D and the block $[0, U_0; U_0^*, -J_d/t - C]$. Remove only the explicit $z - s$ zero rows, and use the positive congruence $\text{diag}(t^{1/2} I_s, t^{1/2} I_{(2d)})$. This leaves $[0, V; V^*, -J_d/t - C]$. Split its last space into $\text{ran } V^*$ and $\ker V$. The paired block $[0, V_r; V_r^*, F]$ has inertia $(s, s, 0)$ and its inverse has zero lower-right block. Therefore its Schur correction to the remaining N -block is exactly zero, and the latter block is $-N^*(J_d + tC)N$. This proves WS41 and accounts for every radical, including persistent zeros.

5.2 The original linear current has at most two interior transition parameters

For $d=1$ there are three exhaustive cases:

- $s=2$: the reduced pencil is empty and $\text{In } H(t) = \text{In } D + (1, 1, z-2)$ for all $t > 0$.
- $s=1$: the reduced pencil is the real scalar $-n^* J_1 n - t \ n^* C n$. Its sign has at most one zero; when both coefficients vanish it is an explicitly retained permanent zero.
- $s=0$: the reduced matrix is $-J_1 - tC$. Its determinant is the real quadratic

$$\det(J_1 + tC) = -1 + t \text{tr}(\text{adj}(J_1)C) + t^2 \det C. \quad (\text{WS42})$$

Its constant coefficient is -1 , so it is not the zero polynomial. There are at most two positive roots. Between them the complete signs follow from its determinant and trace; at each root the zero multiplicity is retained. The signs at $\alpha=0$ and $\alpha=1$ are the actual endpoint inertias, not a limit with zeros discarded.

Thus an entire full-rank covariance activation has at most two interior sign-transition parameters for the canonical linear-current family. The native joint endpoint's coefficients in WS41 are actual matrices; no numerical inertia for that endpoint is assigned merely from its positive covariance.

5.3 A same-map positive-source example attaining two transitions

On C^4 use

$$\begin{aligned} M &= \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \quad \Lambda = [I_2, 0], \\ D_0 &= \text{diag}(1/2, -1/2), \quad D_1 = \text{diag}(-1/2, 3/4), \\ C_0 &= \begin{pmatrix} I_2 & iD_0 \\ -iD_0 & I_2 \end{pmatrix}, \\ C_1 &= \begin{pmatrix} 3I_2 & iD_1 \\ -iD_1 & 3I_2 \end{pmatrix}. \end{aligned} \quad (\text{WS43})$$

Both covariances and their difference are positive definite. Their exact two-by-two blocks have eigenvalues $1 \pm 1/2$, $3 \pm 1/2$, $3 \pm 3/4$, and 2 ± 1 , $2 \pm 5/4$, respectively. Thus the interpolation is a genuine positive added source covariance on the same M and Λ .

The actual quotient metric is $Q_B, \alpha = (1 + 2\alpha)^{-1} I$, and

$$\Lambda S_\alpha \Lambda^* = \text{diag}(1 - 2\alpha, -1 + 5\alpha/2). \quad (\text{WS44})$$

The complete observed signatures are $(1, 1, 0)$ before $\alpha=2/5$, $(1, 0, 1)$ at $2/5$, $(2, 0, 0)$ between $2/5$ and $1/2$, $(1, 0, 1)$ at $1/2$, and $(1, 1, 0)$ afterwards. Both endpoints are indefinite and an interior interval is positive definite. This example evaluates actual source-minimum matrices and has a rank-two observed endpoint current, the exact hypothesis of the finite update reduction. It is not claimed to be the native scalar-source endpoint or a native-period evaluation.

For a covariance error and action error in fixed coefficient coordinates, the exact dual difference is

$$i[\delta T C + T \delta C + \delta T \delta C - \delta C T^* - C \delta T^* - \delta C \delta T^*].$$

Its norm is at most $2(|\delta T| + |C| + |T| + |\delta C| + |\delta T| + |\delta C|)$. Observed errors additionally retain the exact left and right Λ factors and their perturbations. This supplies a finite sign margin, rather than presuming that a nearly zero eigenvalue has a fixed sign.

6. The actual full evolution at every positive exponential time scale

The source CE9-23 proves its joint estimate for the full measured and intrinsic compressed propagators on $b > \psi(0)/2$. The following contour calculation extends the full original propagator and its measured norm to **every fixed $b > 0$** . It does not assign the bounded spectrum of M to its noninvariant compression.

Let, in the original finite metric,

$$M = C + R_0, \quad C = C^\dagger, \quad \text{rank } R_0 = 1, \quad \|C\| \leq c, \\ \|R_0\| = \epsilon, \quad \text{spec } M \subset \{|z| \leq r_0\}.$$

The full original M has $r_0 = k \sqrt{\delta^2 + \gamma^2}$, including its algebraic multiplicities. Put

$$R = 4q \max(1, c, r_0). \quad (\text{WS45})$$

On $|z| = R$, factor each characteristic polynomial into its q actual eigenvalues, with multiplicity. The exact determinant lemma and the rank-one inverse formula give

$$(zI - M)^{-1} - (zI - C)^{-1} = \frac{(zI - C)^{-1} R_0 (zI - C)^{-1}}{\det(zI - M) / \det(zI - C)}. \quad (\text{WS46})$$

Because C is selfadjoint, its resolvent norm is at most $1/(R - c)$. Also

$$\left| \frac{\det(zI - C)}{\det(zI - M)} \right| \leq \left(\frac{1 + 1/(4q)}{1 - 1/(4q)} \right)^q < 2.$$

Indeed its logarithm is at most $1/4 + 1/[4(1 - 1/(4q))] \leq 7/12 < \log 2$; the last strict inequality follows, for example, from the positive series of $\log 2$. Therefore

$$\|(zI - M)^{-1} - (zI - C)^{-1}\| \leq 4\epsilon/R^2.$$

Cauchy's finite-matrix functional calculus, on the same circle enclosing both spectra, gives for every integer $j \geq 1$

$$\|M^j - C^j\| \leq 4\epsilon R^{j-1}. \quad (\text{WS47})$$

The exact $j=1$ difference is R_0 . Sum the absolutely convergent exponential series from $j=2$ onwards:

$$\boxed{\begin{aligned} \|e^{itM} - e^{itC} - itR_0\| &\leq \frac{4\epsilon}{R}(e^{R|t|} - 1 - R|t|) \\ &\leq 2\epsilon R t^2 e^{R|t|}. \end{aligned}} \quad (\text{WS48})$$

For real signed t , $\|e^{itC} - I\| \leq c|t|$, so

$$\boxed{\frac{\|e^{itM} - I - itR_0\|}{1 + |t|\epsilon} \leq c|t| + 2R|t|e^{R|t|}.} \quad (\text{WS49})$$

No commutation of C and R_0 is used. No lower bound on root separation or diagonalizability of M is needed; the determinant factors retain repeated roots.

On the native source WS5-8, R_0 is nilpotent and $\log R = 0_h(k + \log q)$. For $t = \pm \exp(-bq)$ with b in a fixed compact subset of $(0, \infty)$, WS49 is exponentially small. The nilpotent rank-one shear has exact norm

$$S(x) = \frac{\sqrt{x^2 + 4} + x}{2}, \quad x = |t|\epsilon.$$

It has one reciprocal pair of singular values, with all other singular values one. Thus the full original propagator satisfies

$$\boxed{\log \|e^{\pm i e^{-bq} M}\|_G = q(\psi(t_N) - b)_+ + O_{h, \varpi, b}(k \log q),}$$

$$\boxed{\log \text{cond}_G(e^{ie^{-bq}M}) = 2q(\psi(t_N) - b)_+ + O_{h,\varpi,b}(k \log q),} \quad (\text{WS50})$$

$$b > 0.$$

The inverse in the second formula is exactly the full exponential with negative signed time.

For the original measured propagator $U(t) = J_B^\dagger e^{(itM)} J_B$, put $R_B = J_B^\dagger R_0 J_B$, $\epsilon_B = \|R_B\|$ and $h = |\text{tr } R_B|/\epsilon_B$. WS7-8 prove $\epsilon_B \geq \epsilon \sqrt{\Theta_N}$ and $h \leq \sqrt{\alpha_N/\Theta_N}$. Contract WS48 by the actual J_B ; its relative error to $1 + |t|\epsilon_B$ is at most

$$\delta_B = \Theta_N^{-1/2} [c|t| + 2R|t|e^{R|t|}].$$

An exact unitary frame in the two-dimensional range/source span puts R_B/ϵ_B at distance at most $2h$ from the nilpotent unit shear: its matrix is $[\zeta, \sqrt{1-|\zeta|^2}; 0, 0]$, $|\zeta| = h$, after fixing the actual second-vector phase. Therefore

$$\left| \frac{\|U(\pm t)\|_{Q_B}}{S(|t|\epsilon_B)} - 1 \right| \leq 2\delta_B + 2h. \quad (\text{WS51})$$

The finite guard that the right side is below one holds eventually for every fixed $b > 0$, since both Θ_N and the leakage α_N retain WS8. This proves

$$\boxed{\log \|\Lambda e^{\pm ie^{-bq}M} L\|_{Q_B} = q(\psi(t_N) - b)_+ + O_{h,\varpi,b}(k \log q),} \quad (\text{WS52})$$

$$b > 0.$$

In particular its entire original four-return is

$$\begin{aligned} & \mathcal{R} \log \|\Lambda e^{ie^{-bq}M} L\|_{Q_B} \\ &= 2q[(\psi(0) - b)_+ - (\psi(1) - b)_+] + O(k \log q). \end{aligned} \quad (\text{WS53})$$

For every fixed $0 < b < \psi(1)$, the evaluated coefficient is $2[\psi(0) - \psi(1)]$, the plateau formerly outside CE's joint b -domain. The full-space condition number has twice that coefficient. For the physical action $\text{mathscr{A}} = kI/2 + iM$, the scalar $\exp(kt/2)$ is kept at each cutoff and cancels exactly under the four signs and from condition numbers.

No intrinsic inverse is assigned to the measured operator. The original exact defect remains

$$U(-t)U(t) = I - J_B^\dagger e^{-itM}(I - P)e^{itM}J_B. \quad (\text{WS54})$$

The intrinsic operator $V(t) = \exp(it \Lambda M L)$ retains its existing CE proof and its domain $b > \psi(0)/2$. Extending WS52 to the intrinsic small- b domain would require its own spectral estimate: the determinant ratio in WS46 uses the characteristic polynomial of the full M , not that of its compression. The full hidden Volterra equation CE27 and all its cross terms remain unchanged.

7. Source-parameter families, exact signs, and marked classes

The whole-space signature WS15 holds at every positive scalar source measure in the stated moment class, including real exponential tilts within their actual moment domain. For the original tilt $\exp(\theta y) d\mu(y)$, the source isometry is multiplication by $\exp(\theta y/2)$ into the original L_2 space, restricted to the stated polynomial fibres; it does not assert that the changed fibre remains the same unweighted polynomial subspace. Applying WS9-16 to its actual orthogonal polynomials and attained metric proves the same inertia at every tilt. WS39-44 instead treat a positive covariance addition through its exact affine current map. These two family constructions are related

by the common covariance $J_N H_{\text{source}}^{(-1)} J_N^*$; no equality between a mixed covariance and an unspecified moment Gram is presumed.

On each original eigenclass $Mx = \omega x$, the full linear current is exactly

$$\langle x, W_y x \rangle_G = -2\Im \omega \|x\|_G^2. \quad (\text{WS55})$$

For the stipulated terminal $\omega = k \gamma - i k \delta$ this is positive. It does not follow that a span of several same-half-plane eigenclasses is a positive subspace: in an eigenclass frame x_j the exact current Gram is

$$\langle x_i, W_y x_j \rangle_G = i(\omega_j - \bar{\omega}_i) \langle x_i, x_j \rangle_G. \quad (\text{WS56})$$

All off-diagonal source Grams remain. The exact auxiliary Gaussian quotient $Q = (y^2 + 1)(y^2 + 4)$, $N=3$, has positive individual current values on the $-i$ and $-2i$ eigenlines but a signature- $(1,1)$ current on their span. The checker constructs the actual eigenvectors and verifies that statement. This is consistent with, and illustrates the necessity of, the whole-space signature WS15.

For the prescribed observed terminal class, the original TR20 formula remains

$$j_N = 2\epsilon \sqrt{\det C_B \Im(\bar{c}_g c_h)} - 2\epsilon(\Im r) |c_g|^2. \quad (\text{WS57})$$

The new source proves the last term exponentially small in absolute value, with its observed-norm guard, and proves an exponentially small coherence deficit. Neither property assigns the sign of the retained imaginary product. This continuation does not rotate that marked class or replace its observation to assign a sign. It completes whole-space, polynomial-word and mixed-form classifications instead.

8. A necessary correction to the new denominator return

The original source profiles are different scalar functions connected by RC10–12. Retain exactly

$$\begin{aligned} a_0 &= \psi(0) = \log(4/\pi), \quad a_1 = \psi(1) = \log \frac{\kappa_1(1+r_1)}{E(\kappa_1)^2}, \\ E(\kappa_1) &= 2r_1 K(\kappa_1), \\ J(t) &= 4 \int_0^t \psi(s) ds - 2(1+t)\psi(t) + 2a_0, \quad C_\partial = 2J(1). \end{aligned} \quad (\text{WS58})$$

RD10–14 correctly give $\log \alpha_{\text{resp}} = 2q \psi(t_N) + 0(k \log q)$. Its first equality in RD15 therefore correctly yields

$$\boxed{\mathcal{R} \log \alpha_{\text{resp}} = 4(a_0 - a_1)q + O(k \log q)}. \quad (\text{WS59})$$

Read as a new local definition, RD15 assigns its symbol C_{boundary} a different number; that local label cannot be propagated into the existing CK/PT kernel receivers. Keeping the established global $C_{\text{boundary}} = 2J(1)$, the following equality to $2C_{\text{boundary}} q$ in RD15 is false. Indeed WS58 gives

$$2C_\partial - 4(a_0 - a_1) = 16 \int_0^1 \psi(s) ds - 12a_1 + 4a_0 \geq 4(a_0 + a_1) > 0, \quad (\text{WS60})$$

because the stated ψ is positive decreasing. This is an exact derivation of the discrepancy, independent of a numerical resemblance. OCP46–47 already give the correct half-size linear-current coefficient $2(a_0 - a_1)$.

The companion outward integer checker certifies

$$0.35009104650043305003462290080515 < 2(a_0 - a_1) < 0.35009104650043305003462290080516,$$

$$0.70018209300086610006924580161031 < 4(a_0 - a_1) < 0.70018209300086610006924580161032.$$

(WS61)

In contrast `2C_boundary` lies between `2.70856397565058426576797955030769` and `2.70856397565058426576797955030770`. The intervals are disjoint. The current profile, the valid denominator pointwise bounds, and the earlier kernel-volume coefficient are unchanged; only RD15's identification of two constants is corrected.

The checker uses denominator 10^{80} , outward integer addition/multiplication/division and square roots, a Machin enclosure of π , 4096 normalized elliptic-series terms with geometric tails, 125 certified bisection decisions, and a 160-term logarithm series with its full remainder. Its numerical scope is the universal scalar constants, not a native period or an off-line zero.

9. Verification and limits of the result

`check_whole_space_identities.py` tests the full source minima, every polynomial boundary factor, exact signatures, positive/negative/isotropic witnesses, real repeated-primary relations, real divisor words, the full relation Gram with its off-diagonal blocks, signed nonzero products, actual exterior metrics and additive actions, same-map source activation, singular endpoint-current radicals, and original mixed phase orbits. Its negative controls retain the precise failures of a real-root sign-changing relation, a complex divisor word, inference of positive subspaces from eigenline signs, and omission of the eigenclass defect under a hidden-phase rotation. Counts and executed modes are recorded by the package verification manifest, not asserted from the proof alone.

`check_rational_currents.py` verifies the full denominator congruence, the Bezout/Krylov factor, gcd cancellation, and the exact sign counts with repeated primaries and real or repeated denominator poles. `check_reduced_pencils.py` verifies the complete singular-base inertia formula for $d=1,2,3$, every possible null-coupling rank, and three positive rational interpolation weights, including the retained radical. `check_evolution_diagnostics.py` is separately labeled noninterval: it tests the proved contour bound at 140-digit precision on matrices with the exact spectrum $\{i, -i\}$ and very large rank-one boundary terms. Its numerical successes are not proofs of the analytic bounds.

The new finite results are WS15-24 (including the complete rational extension WS21a-e), WS26-29, WS35b-44, and WS46-54, with their scope and maps given above. The large-degree applications WS31-33 and WS50-53 use the retained source estimates RC22-27 and the actual full characteristic polynomial, not a new assumption about a native projection. The exact whole-space signs are determined; the separate marked terminal imaginary product WS57 is not numerically or asymptotically evaluated here. No claimed zero, period, or RH conclusion follows from an auxiliary sign fixture.

Source locations and reading coverage

Read in the pinned GitHub source, with full mathematical bodies used: `OBSERVED_CURRENT_PLANE.tex` OCP1-59 (finite source/conductor/current identities and existing signed-time bound); `CUTOFF_CURRENT_CONNECTION.md` RC1-34 (profiles, finite source costs, current and evolution returns); `EVOLUTION_PROOF.md` CE1-29; `RESPONSE_DENOMINATORS.md` RD1-21; `TERMINAL_COHERENCE_PROOF.md` TR1-24 including TR12a. The last terminal-conductor domain itself is inherited through the TR/RD references, not re-proved by replacing it with a weaker period assumption. The supplied `/mnt/`

data/Native_Cutoff_Profile_20260922/COMPLETE_PROOFS.md was reviewed as the earlier cutoff-profile context; its coefficient and measured-phase results are not relabeled as new in this note.

Human primary reference: NIST DLMF 18.2 for the real orthogonal-polynomial recurrence, positivity and Christoffel conventions. All finite source, inertia, mixed-current and contour formulas needed here are proved in this note. The Gamma and elliptic estimates used for native rates retain the original RC/OCP source citations; the interval checker derives its scalar series and remainders directly. A separately executed finite checker supplements rather than replaces the proofs.

Native adjacent source steps have no whole-space word-sign transitions

22 September 2026. Independent derivation NW1–NW18. For every real polynomial word of degree at most half the original quotient dimension, its full current has the same balanced inertia throughout the actual next-source covariance step. The proof uses the original positive real source, every relation of the full root polynomial, and the actual next monic class. A final exact denominator congruence also settles real Möbius transforms of the original multiplication operator. These results concern full-space currents; their exact map to observed currents is retained below.

1. Original source and currents

Keep the original fixed simple quartet, period, arithmetic source and physical coordinate:

$$q = (k+1)^2, \quad k \geq 17, \quad k \equiv 1 \pmod{4},$$

$$Q(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E = \mathbb{C}[y]/Q, \quad M[f] = [yf]. \quad (\text{NW1})$$

Here $0 < \delta < 1/2$, $\gamma > 2$. In the real monomial coefficient basis $1, y, \dots, y^{q-1}$, M is real and q is even. Pairing the roots with indices $a, k-a$ proves exactly

$$Q(y) = \prod_{b=0}^k \prod_{a=0}^{(k-1)/2} \{[y - (2b-k)\gamma]^2 + (2a-k)^2\delta^2\} > 0 \quad (\text{NW2})$$

$$(y \in \mathbb{R}).$$

In particular $Q(0) > 0$, and $\det M = (-1)^q Q(0) = Q(0) > 0$. The determinant formula follows from the companion matrix or the characteristic polynomial $\det(zI - M) = Q(z)$.

Let p_j be the original real monic orthogonal polynomials for the complete positive arithmetic source $d\mu$, let $\omega_j = \|p_j\|_\mu^2 > 0$, and let $b_j = [p_j]$ be their coefficient columns in E . Every nonzero polynomial has positive squared source norm. The complete quotient covariance at degree $N \geq q-1$ is

$$D_N = G_N^{-1} = \sum_{j=0}^N \frac{b_j b_j^\top}{\omega_j}. \quad (\text{NW3})$$

Indeed the matrix with all columns $b_j/\sqrt{\omega_j}$ is the complete source map. Its kernel consists of all degree-at-most- N multiples of Q . Minimizing its Euclidean coefficient norm on each fibre gives the inverse covariance G_N , as proved in NV12 of the accompanying complete native source-step proof. Thus no relation or lower source component is removed.

Let $p \in \mathbb{R}[y]$ have degree $d \leq q/2$, leading coefficient $c_d \neq 0$, and put $T = p(M)$. Define the real skew matrix $A_N = TD_N - D_N T^\top$, the Hermitian dual current $S_N = iA_N$, and the native addition

$$u = b_{N+1}/\sqrt{\omega_{N+1}}, \quad D(t) = D_N + t u u^\top, \quad G(t) = D(t)^{-1},$$

$$0 \leq t \leq 1.$$

$$S(t) = i[TD(t) - D(t)T^\top] = S_N + it[(Tu)u^\top - u(Tu)^\top]. \quad (\text{NW4})$$

In particular, the theorem applies to both literal original adjacent paths $N = q-1 \rightarrow q$ and $N = 2q-1 \rightarrow 2q$, retaining all four endpoints. The finite proof also holds for every $N \geq q-1$ for which these same complete source moments are defined.

The source is real, so transpose is the complex adjoint of every displayed real matrix. The original full current in source-metric isometric coordinates is

$$G(t)^{-1/2}i[G(t)T - T^\top G(t)]G(t)^{-1/2} = G(t)^{1/2}S(t)G(t)^{1/2}. \quad (\text{NW5})$$

This is an invertible congruence of the exact original current; its inertia agrees with that of $S(t)$. Here inertia means the ordered triple of positive, negative and zero eigenvalue multiplicities. For an invertible congruence this invariance follows by mapping positive and negative subspaces bijectively and applying the maximal dimensions of such subspaces; this is the finite form argument used throughout.

At $d = 0$, T is scalar and every current is zero. Henceforth $1 \leq d \leq q/2$.

2. The complete boundary block

Expand the exact polynomial product $p(y)p_j(y)$ in the original monic basis. Its coefficients satisfy

$$\begin{aligned} p(y)p_j(y) &= \sum_{\ell=0}^{j+d} a_{\ell j} p_\ell(y), \quad \frac{a_{\ell j}}{\omega_j} \\ &= \frac{\langle p p_j, p_\ell \rangle_\mu}{\omega_j \omega_\ell} \\ &= \frac{a_{j\ell}}{\omega_\ell}. \end{aligned}$$

All coefficients are real. By orthogonality $a_{\ell j} = 0$ for $|\ell - j| > d$, and $a_{j+d,j} = c_d$. Substitution in NW3 makes every pair with $j, \ell \leq N$ cancel. The only remaining rows have $N - d + 1 \leq j \leq N$ and $N + 1 \leq \ell \leq N + d$.

Write

$$\begin{aligned} V_N &= [b_{N-d+1}, \dots, b_N, b_{N+1}, \dots, b_{N+d}], \\ H_{rc} &= \frac{a_{N+r, N-d+c}}{\omega_{N-d+c}}, \quad 1 \leq r, c \leq d. \end{aligned}$$

Then

$$\boxed{A_N = V_N B_N V_N^\top, \quad B_N = \begin{pmatrix} 0 & -H^\top \\ H & 0 \end{pmatrix}.} \quad (\text{NW6})$$

The bandwidth gives $H_{rc} = 0$ for $r > c$, so H is upper triangular with

$$H_{rr} = \frac{c_d}{\omega_{N-d+r}}, \quad \det H = \frac{c_d^d}{\prod_{j=N-d+1}^N \omega_j} \neq 0. \quad (\text{NW7})$$

This proves the finite boundary matrix is nonsingular for every degree cutoff; no generic-position assumption is used.

Here is the required consecutive-class independence. For any $\ell \geq 0$ and $1 \leq r \leq q$, a relation $\sum_{j=\ell}^{\ell+r-1} v_j b_j = 0$ gives a polynomial $P = \sum v_j p_j = Qh$ with $\deg h \leq \ell + r - 1 - q \leq \ell - 1$. Orthogonality to h and NW2 yield

$$0 = \langle P, h \rangle_\mu = \int_{\mathbb{R}} Q(y) |h(y)|^2 d\mu(y). \quad (\text{NW8})$$

The integrand is strictly positive away from the zero set of h ; a nonzero h cannot vanish on the source's support because its squared norm is positive. Thus $h = 0$, $P = 0$, and monic triangularity gives every $v_j = 0$. If the degree bound for h is negative, divisibility already implies $P = 0$. The proof applies to complex coefficients as well, with the same Hermitian integral.

In particular V_N has full column rank $2d$. The matrix iB_N is congruent to $i \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$, by changing its second block with $H^{-\top}$. That latter Hermitian matrix has d positive and d negative eigenvalues. Full-column-rank factorization in NW6 then proves

$$\boxed{\begin{aligned} \text{rank } S_N &= 2d, \\ \text{ran } S_N &= \text{span}\{b_{N-d+1}, \dots, b_{N+d}\}, \\ \text{In } S_N &= (d, d, q - 2d). \end{aligned}} \quad (\text{NW9})$$

For rigor concerning the rectangular factor: choose a basis extending the columns of V_N ; its inverse transpose converts $V_N B_N V_N^\top$ into $B_N \oplus 0$. This proves the exact range and the inertia assertion.

3. Degrees strictly below half dimension

Suppose $2d < q$. Then u lies in $\text{ran } V_N$, so write $u = V_N a$ with its unique real coordinate column a . The product $p(y)p_{N+1}(y)$ involves exactly the possible monic degrees $N - d + 1, \dots, N + d + 1$, with top coefficient $c_d \neq 0$. By NW8 these $2d + 1$ consecutive classes are independent. Hence Tu lies outside $\text{ran } V_N$, and $F = [V_N, Tu]$ has full column rank $2d + 1$.

Use this actual final vector Tu as a new coordinate; no orthogonal projection or replacement source direction is introduced. NW4 and NW6 give

$$\boxed{S(t) = iF \begin{pmatrix} B_N & -ta \\ ta^\top & 0 \end{pmatrix} F^\top.} \quad (\text{NW10})$$

Since B_N^{-1} is real skew, $a^\top B_N^{-1} a = 0$. Put $h = tB_N^{-1} a$ and $R_t = \begin{pmatrix} I & h \\ 0 & 1 \end{pmatrix}$. Direct multiplication gives

$$\begin{aligned} R_t^\top \begin{pmatrix} B_N & -ta \\ ta^\top & 0 \end{pmatrix} R_t &= \\ \begin{pmatrix} B_N & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{NW11})$$

Indeed $B_N h = ta$, $h^\top B_N = -ta^\top$, and the last scalar is $t^2 a^\top B_N^{-1} a = 0$. Both coordinate maps are explicit and invertible on their indicated ranges. Therefore

$$\boxed{\text{In } S(t) = (d, d, q - 2d) \quad \text{for every real } t, \quad 2d < q.} \quad (\text{NW12})$$

The original positive covariance interpretation is used on $0 \leq t \leq 1$; the displayed algebraic inertia identity itself holds for all real t . At no positive interior time does a whole-space current eigenvalue cross zero. The moving support and kernel are retained: in the coefficient matrix of NW10 its one-dimensional kernel is spanned by $(tB_N^{-1} a, 1)$, and the original full kernel is the inverse image of this line under $F^\top$. Thus the nullspace moves even though its dimension and the positive/negative counts stay fixed.

This also verifies the vanishing scalar in a bordered update argument without relying on a pseudoinverse identification. In the original coefficient basis, S_N^+ is purely imaginary and skew under transpose, so $u^* S_N^+ u = 0$ for real u ; NW11 proves the stronger complete congruence directly.

4. Half degree and Pfaffian orientation

Suppose $2d = q$, so V_N is square and nonsingular. Its determinant has a fixed positive sign. To prove this without assuming any orientation of a generic source matrix, introduce only for this proof the pushforward measure μ_ε under $y = \varepsilon x$, for $\varepsilon > 0$. Its monic

orthogonal polynomials are $p_j^{(\varepsilon)}(y) = \varepsilon^j p_j(y/\varepsilon)$. Their orthogonality follows by substitution in the integral, and all nonzero polynomials retain positive norm. Keep the original Q fixed. The positivity proof NW8 therefore shows that the q -column matrix of their classes at degrees $\ell, \dots, \ell + q - 1$ is nonsingular for every $\varepsilon > 0$.

Every entry is a polynomial in ε , and at $\varepsilon = 0$ the columns become the original monomial classes $[y^\ell], \dots, [y^{\ell+q-1}]$. This limiting matrix is M^ℓ in the stated monomial coefficient basis. Hence its determinant is $(\det M)^\ell > 0$. Continuity and nonvanishing on $\varepsilon > 0$ prove, at $\varepsilon = 1$,

$$\boxed{\det[b_\ell, \dots, b_{\ell+q-1}] > 0 \quad (\ell \geq 0).} \quad (\text{NW13})$$

The scaled measure is an auxiliary orientation proof with its exact pullback given above; it does not replace any original source metric in NW3–12.

Define the Pfaffian of a real $2d$ -dimensional skew matrix A by $\Omega_A^d/d! = \text{Pf}(A)e_1 \wedge \dots \wedge e_{2d}$, where $\Omega_A = \sum_{i < j} A_{ij}e_i \wedge e_j$. Exterior expansion proves $\text{Pf}(VAV^T) = \det V \text{Pf}(A)$. For NW6, taking the d cross-block terms in that expansion gives

$$\text{Pf}(B_N) = (-1)^{d(d+1)/2} \det H,$$

$$\boxed{\text{Pf}(A_N) = (-1)^{d(d+1)/2} \frac{c_d^d \det V_N}{\prod_{j=N-d+1}^N \omega_j}.} \quad (\text{NW14})$$

The sign can also be checked at $d = 1$: the first upper entry is $-H_{11}$, fixing the convention. All factors other than the explicitly displayed fixed sign and c_d^d are positive. Thus the endpoint Pfaffians at $N, N + 1$ have the same nonzero sign.

The update $A(t) - A_N = t[(Tu)u^T - u(Tu)^T]$ has decomposable two-form. Its exterior square is zero, so the Pfaffian is affine in t . At the actual endpoint, NW3–4 give $A(1) = A_{N+1}$ for the same polynomial p . Therefore

$$\boxed{\begin{aligned} \text{Pf}(A(t)) &= (1-t) \text{Pf}(A_N) + t \text{Pf}(A_{N+1}) \neq 0 \\ &\quad (0 \leq t \leq 1). \end{aligned}} \quad (\text{NW15})$$

A skew matrix with a nonzero Pfaffian is nonsingular: a kernel vector, extended to a basis, would make its corresponding row and column zero after congruence and force the top exterior coefficient to vanish. A real skew matrix has opposite pairs of nonzero eigenvalues after multiplication by i : complex conjugation sends an eigenvector of iA at real eigenvalue λ to an eigenvector at $-\lambda$. Hence

$$\boxed{\text{In } S(t) = (q/2, q/2, 0) \quad (0 \leq t \leq 1).} \quad (\text{NW16})$$

No half-degree native whole-space transition occurs. Quantitatively its Pfaffian ratio is exactly

$$\begin{aligned} \frac{\text{Pf}(A_{N+1})}{\text{Pf}(A_N)} &= \frac{\det V_{N+1}}{\det V_N} \\ \frac{\omega_{N-d+1}}{\omega_{N+1}} &> 0. \end{aligned}$$

This retains the complete consecutive-source determinants instead of replacing the endpoint sign by a continuity guess.

5. Exact rational receiver and observed maps

A real Möbius word in the original multiplication operator is settled by an exact denominator congruence. Let real a, b, c, d_0 satisfy $(c, d_0) \neq (0, 0)$ and put $R = cM + d_0I$. This R is automatically invertible: if $c \neq 0$, an eigenvalue $-d_0/c$ would be a real root

of Q , excluded by NW2; if $c = 0$, it is a nonzero scalar. Define $T_{\text{mob}} = (aM + bI)R^{-1}$. Multiplication, using commutativity of both polynomials in M , proves

$$\boxed{R i[T_{\text{mob}}D(t) - D(t)T_{\text{mob}}^T]R^T = (ad_0 - bc) i[MD(t) - D(t)M^T].} \quad (\text{NW17})$$

Indeed expand $(aM + bI)D(cM + d_0I)^T - (cM + d_0I)D(aM + bI)^T$; the MDM^T and D terms cancel and the other two have the displayed determinant coefficient. If $ad_0 - bc \neq 0$, NW12 for degree one gives inertia $(1, 1, q - 2)$ at every native step point; multiplying a balanced inertia by a negative scalar preserves the triple. If $ad_0 - bc = 0$, the numerator is a real scalar multiple of R , the word is constant, and its current is zero. This settles this rational family completely. No general rational-function activation assertion is made.

Finally retain the original observation Λ , its moving complete covariance metric $Q_B(t) = (\Lambda D(t)\Lambda^*)^{-1}$, and minimum section $L(t) = D(t)\Lambda^*Q_B(t)$. The observed word current is exactly

$$\boxed{iQ_B(t)\Lambda[TD(t) - D(t)T^T]\Lambda^*Q_B(t) = Q_B(t)\Lambda S(t)\Lambda^*Q_B(t).} \quad (\text{NW18})$$

Thus the full current reaches the observed one by the indicated compression and positive metric congruence. NW12 and NW16 do not assign the inertia of that compression or a marked vector's sign. The source-step identities NV1–26 supply their exact separate receiving maps. The no-transition theorem here applies to the entire original source space along the actual native source addition; it does not contradict activation formulas for other source columns.

6. Sources and finite verification

The incoming WS39–44 full affine-activation proof was read at lines 508–601. It includes singular bordered updates and a positive-source example with observed transitions; it does not evaluate the special native next-monic column proved here. The independent WD1–20 source and word-sign proof was also read, including the attained minimum, complete boundary factor and positivity independence. NW6–9 supplies the corresponding dual-coordinate factor directly; NW10–16 then proves its complete native adjacent-step continuation. Both previously unpublished incoming sources accompany the successor rather than receiving invented public links.

The complete original minimum covariance and source parity used above are proved in the accompanying NV12–13 and publicly in OCP5–7. The original polynomial and complete metric are RC1–2. These references preserve the human Gamma/orthogonal-polynomial source citations, including T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF18.22.8. The positive-source independence, full boundary factorization, native congruence and Pfaffian orientation are proved in full here.

The auxiliary checker retains all source covariance rows in a four-root Gamma quotient and tests degrees one and two, the literal low/high adjacent paths, the exact native skew factorization, the Pfaffian interpolation and its endpoint signs. It separately tests the rectangular congruence and Möbius denominator formula. These are finite algebraic checks, not numerical data for hypothetical zeta zeros or replacements for the original five-orbit observation. The analytic proof above covers the stated original degree range without relying on those finite examples.

The original adjacent source step and its marked-current correction

22 September 2026. Independent verification and continuation NV1-26. The original source covariance, observation and complete minimum section are retained throughout. The finite interpolation identities first hold for every rank-one source addition. Applying the actual next monic source column then gives the stronger terminal-mass estimate and its original cutoff profile. No current sign is assigned without its actual endpoint values and trace change.

1. The exact minimum-map interpolation

Work in the fixed original coefficient space $E = \mathbb{C}[y]/Q_k$, with its full degree- N minimum metric $G_0 = G_N$, covariance $D_0 = G_0^{-1}$, and unchanged observation $\Lambda : E \rightarrow B$. For a source column $u \in E$, set

$$\begin{aligned} D(t) &= D_0 + tuu^*, & G(t) &= D(t)^{-1}, & Q(t) &= (\Lambda D(t) \Lambda^*)^{-1}, \\ L(t) &= D(t) \Lambda^* Q(t), & 0 \leq t &\leq 1. \end{aligned} \quad (\text{NV1})$$

The source covariance is positive definite for every such t , and Λ is surjective onto its actual image B ; hence every inverse in NV1 exists. Put $Q_0 = Q(0)$, $L_0 = L(0)$, $P_0 = L_0 \Lambda$, $u_B = P_0 u$, $u_K = (I - P_0)u$, and $z = \Lambda u$. These are the original G_0 -orthogonal observed and kernel components. For a fixed observation ξ , define

$$\begin{aligned} \beta &= z^* Q_0 z = \|u_B\|_{G_0}^2, & \rho &= z^* Q_0 \xi, & w &= |\rho|^2, & Y_0 &= \xi^* Q_0 \xi, \\ s &= \frac{t}{1 + \beta t}. \end{aligned} \quad (\text{NV2})$$

Multiplication of the proposed inverse by $\Lambda D(t) \Lambda^*$ proves the rank-one inverse formula. Substitution into $L(t)$, retaining $u = u_B + u_K$, then proves

$$Q(t) = Q_0 - s Q_0 z z^* Q_0, \quad L(t) = L_0 + s u_K z^* Q_0. \quad (\text{NV3})$$

Indeed the coefficient of u is $t - st\beta = s$, and the remaining subtraction is $s u_B z^* Q_0$. Thus

$$L(t) \xi = L_0 \xi + s \rho u_K, \quad \xi^* Q(t) \xi = Y_0 - s w. \quad (\text{NV4})$$

These are exact maps of the original fibres. They include the entire original kernel; u_K is not an independently chosen direction.

Let $M[p] = [yp]$ remain fixed and write

$$M_B(t) = \Lambda M L(t), \quad \Phi(t) = i \xi^* Q(t) \Lambda [M D(t) - D(t) M^*] \Lambda^* Q(t) \xi,$$

$$\tau(t) = i \operatorname{tr} \{ Q(t) \Lambda [M D(t) - D(t) M^*] \Lambda^* \}. \quad (\text{NV5})$$

Both currents are real. Cyclicity of trace and $D(t) \Lambda^* Q(t) = L(t)$ give $\Phi(t) = -2\Im[\xi^* Q(t) M_B(t) \xi]$ and $\tau(t) = -2\Im \operatorname{tr} M_B(t)$. Put $a_K = \Lambda M u_K$. NV3 implies

$$M_B(t) = M_B(0) + s a_K z^* Q_0, \quad \tau(t) = \tau_0 - 2s \Im \langle u_B, M u_K \rangle_{G_0}.$$

Consequently, with $\Delta\tau = \tau_1 - \tau_0$,

$$\boxed{\begin{aligned} \Delta\tau &= -\frac{2}{1 + \beta} \Im \langle u_B, M u_K \rangle_{G_0}, \\ \tau(t) &= \frac{(1 - t)\tau_0 + t(1 + \beta)\tau_1}{1 + \beta t}. \end{aligned}} \quad (\text{NV6})$$

The sign comes from $i(A - \bar{A}) = -2\Im A$. This proof holds for every source column, including a column with a nonzero kernel component.

2. The marked current and its exact quadratic receiver

Expand $\xi^* Q(t) M_B(t) \xi$ using NV3:

$$\xi^* Q_0 M_B(0) \xi + s[\rho \xi^* Q_0 a_K - \bar{\rho} z^* Q_0 M_B(0) \xi] - s^2 w z^* Q_0 a_K.$$

Taking $-2\Im$ shows that $\Phi(t) = q_0 + q_1 s + q_2 s^2$, with

$$q_0 = \Phi_0, \quad q_2 = 2w\Im\langle u_B, M u_K \rangle_{G_0} = -(1 + \beta)w\Delta\tau.$$

Since $s(1) = 1/(1 + \beta)$, the endpoint Φ_1 determines the remaining coefficient. The exact receiver into SC17-18 is therefore

$$\begin{aligned} q_0 &= \Phi_0, \\ q_1 &= (1 + \beta)(\Phi_1 - \Phi_0) + w\Delta\tau, \\ q_2 &= -(1 + \beta)w\Delta\tau. \end{aligned} \tag{NV7}$$

For the terminal eigenclass these are precisely SC17's coefficients, because the complete source step and u_B, u_K, ρ are identical. The derivation above also proves the polynomial identity without an eigenclass hypothesis.

Returning to the original source coordinate t gives

$$\Phi(t) = \frac{(1 - t)\Phi_0 + t(1 + \beta)\Phi_1}{1 + \beta t} + \frac{t(1 - t)w\Delta\tau}{(1 + \beta t)^2}. \tag{NV8}$$

Thus the additional marked-current term is determined by the actual trace change and the observed source overlap. It does not disappear when the minimum section moves.

The coordinate

$$\lambda = (1 + \beta)s = \frac{(1 + \beta)t}{1 + \beta t}$$

is a strictly increasing bijection of $[0, 1]$, with inverse $t = \lambda/(1 + \beta - \beta\lambda)$. In this coordinate

$$\begin{aligned} \tau &= (1 - \lambda)\tau_0 + \lambda\tau_1, \\ \Phi &= (1 - \lambda)\Phi_0 + \lambda\Phi_1 + \lambda(1 - \lambda)\frac{w\Delta\tau}{1 + \beta}. \end{aligned} \tag{NV9}$$

This is an exact change of the interpolation parameter, with its inverse supplied; no original metric or source column has been changed.

Since $\lambda(1 - \lambda)$ has maximum $1/4$, the absolute deviation from the weighted endpoint chord is

$$\begin{aligned} \max_{0 \leq t \leq 1} \left| \Phi(t) - \frac{(1 - t)\Phi_0 + t(1 + \beta)\Phi_1}{1 + \beta t} \right| \\ = \frac{w|\Delta\tau|}{4(1 + \beta)}. \end{aligned} \tag{NV10}$$

It is attained at $\lambda = 1/2$, equivalently $t = 1/(\beta + 2)$. If $w\Delta\tau = 0$, the deviation is zero throughout and this point is still a maximizer. The signed deviation at that point is $w\Delta\tau/[4(1 + \beta)]$. Cauchy-Schwarz in the original Q_0 metric proves

$$\begin{aligned} w &\leq \beta Y_0, \quad \frac{w|\Delta\tau|}{4(1 + \beta)} \\ &\leq \frac{\beta}{1 + \beta} \frac{Y_0|\Delta\tau|}{4}. \end{aligned} \tag{NV11}$$

This retains a factor stronger than the bound $Y_0|\Delta\tau|/4$.

If $\beta = 0$, positivity of Q_0 implies $z = 0$, so $w = 0$, $u_B = 0$, and NV3 gives $Q(t) = Q_0$, $L(t) = L_0$. Both currents are constant and NV6-11 hold without division by β . Likewise $\xi = 0$ makes all marked currents and w, Y_0 zero. No normalized terminal overlap is used in either edge case.

The sign of $\Delta\tau$ also has a finite consequence: the current polynomial in λ is concave for $\Delta\tau > 0, w > 0$, and convex for $\Delta\tau < 0, w > 0$. Its second derivative is exactly $-2w\Delta\tau/(1 + \beta)$. The three-value SC17 criterion is preserved. In particular positive endpoints remain positive when the trace increases, and negative endpoints remain negative when the trace decreases. The other cases require the actual vertex or its bound NV10; no endpoint sign is inferred.

3. The original next-source column and recurrence

Now specialize to the actual source addition, without substituting a different column:

$$\begin{aligned} u &= \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}, \quad b_j = [p_j], \\ D(t) &= G_N^{-1} + t \frac{b_{N+1}b_{N+1}^*}{\omega_{N+1}}. \end{aligned} \tag{NV12}$$

For completeness, let $V_N : \mathbb{C}^{N+1} \rightarrow E$ have all columns $b_j/\sqrt{\omega_j}$, $0 \leq j \leq N$. The domain norm is exactly the coefficient norm in the original orthonormal polynomial basis. Its kernel consists of every degree-at-most- N polynomial divisible by Q_k ; these are the complete original relations. Since $N \geq q-1$, the map is surjective. For an original class x , its minimum coefficient vector is $V_N^*(V_N V_N^*)^{-1}x$: it maps to x , is orthogonal to $\ker V_N$, and subtracting it from any other preimage leaves a vector in that kernel. Its squared norm is $x^*(V_N V_N^*)^{-1}x$. Thus $G_N = (V_N V_N^*)^{-1}$, with no omitted relation. Adding the one next column proves $D(1) = G_{N+1}^{-1}$ in NV12.

The two native adjacent paths used for the original four-cutoff return are $N = q-1 \rightarrow q$ and $N = 2q-1 \rightarrow 2q$. Their endpoint source and terminal estimates are therefore entirely within the four original cutoffs. The finite identities apply wherever the same complete source construction is defined; the quantitative terminal statements here are asserted on these two paths.

The original even source has the exact monic recurrence

$$Mb_{N+1} = b_{N+2} + \frac{\omega_{N+1}}{\omega_N} b_N.$$

The zero middle coefficient follows by integrating the odd function $y|p_{N+1}(y)|^2$ against the even source; the lower coefficient follows by testing against p_N , using the leading coefficient one in the reverse recurrence. The relation-quotient map carries this polynomial identity to the same classes b_j .

Let $z_j = \Lambda b_{N+j}$, $j = 0, 1, 2$. The original boundary decomposition OCP5-7 gives

$$MD_0 - D_0 M^* = \frac{b_{N+1}b_N^* - b_N b_{N+1}^*}{\omega_N}.$$

Insert NV12 and the exact recurrence. Its b_N, b_{N+1} contribution subtracts t times this boundary pair. Thus the complete unmetricized observed skew current is

$$\boxed{\begin{aligned} \Lambda[MD(t) - D(t)M^*]\Lambda^* &= (1-t) \frac{z_1 z_0^* - z_0 z_1^*}{\omega_N} \\ &\quad + t \frac{z_2 z_1^* - z_1 z_2^*}{\omega_{N+1}}. \end{aligned}} \tag{NV13}$$

All signs are fixed by the displayed order. Both endpoints share the actual next-source column z_1 . In the native edge case $\beta = 0$, $z_1 = 0$, so NV13 vanishes and both observed currents are zero, strengthening their general constancy.

The original source identity and its map to the quotient are fully retained in the cited OCP proof. Human orthogonal-polynomial provenance is T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF Chapter18, with the Gamma comparison and its original recurrence read as 18.22.8. The recurrence for the arithmetic measure used here has just been derived from its parity and monic orthogonality; it is not asserted to be the Gamma recurrence itself.

4. The actual terminal mass controls the peak correction

Retain the full terminal domain of RC35-44: the original simple quartet, admitted five-orbit period with the stated finite corner radius, and $\gamma \geq 1000\delta$. Take the same terminal cardinal x , $\xi = \Lambda x$. Then $Y_0 > 0$. In the actual source isometry $J = L_0 Q_0^{-1/2}$, retain ST5 and ST13-14:

$$\begin{aligned} e &= b_N/\sqrt{E_N}, \quad f = b_{N+1}/\sqrt{F_N}, \quad u_e = J^\dagger e, \quad v_f = J^\dagger f, \\ a &= \|u_e\|^2, \quad d = \|v_f\|^2, \quad r = u_e^* v_f, \quad \mathfrak{d} = ad - |r|^2, \quad g = u_e/\sqrt{a}, \\ h &= (v_f - (r/a)u_e)/\sqrt{\mathfrak{d}/a}. \\ \hat{y} &= Q_0^{1/2}\xi/\sqrt{Y_0}, \quad c_g = g^*\hat{y}, \quad c_h = h^*\hat{y}, \\ \mu &= |c_g|^2 + |c_h|^2, \quad \mathfrak{s} = \hat{y}^*(u_e u_e^* + v_f v_f^*)\hat{y}. \end{aligned} \tag{NV14}$$

The support parameter μ is the combined mass in the actual two current directions; components in their orthogonal complement remain in the unit norm of $\hat{y}$. The complete area certificate gives $\mathfrak{d} \geq \Theta_N > 0$, so $d > 0$. The phase-coherent terminal equation gives $|c_g| \leq \eta_N$, with the exact finite η_N in TR10.

Since $Q_0^{1/2}z_1 = \sqrt{F_N}v_f$, the native source parameters are exactly

$$\begin{aligned} \beta &= \frac{F_N}{\omega_{N+1}}d, \quad w = \beta Y_0 \rho_f, \\ \rho_f &:= \frac{|v_f^* \hat{y}|^2}{d}. \end{aligned} \tag{NV15}$$

Thus the marked source overlap is not replaced by its largest possible value.

The unit vector $v_f/\sqrt{d}$ has coordinates $(r, \sqrt{\mathfrak{d}})/\sqrt{ad}$ in the g, h plane. Its orthogonal unit complement has coordinates $(\sqrt{\mathfrak{d}}, -\bar{r})/\sqrt{ad}$. Taking the squared component of $\hat{y}$ on this complement proves

$$\begin{aligned} 0 \leq \mu - \rho_f &= \frac{|\sqrt{\mathfrak{d}}c_g - rc_h|^2}{ad} \\ &\leq E_f := \frac{(\sqrt{\mathfrak{d}}\eta_N + |r|)^2}{ad}. \end{aligned} \tag{NV16}$$

The complex conjugations are fixed by the two displayed unit vectors; replacing rc_h by $\bar{r}c_h$ generally changes the quantity. Because $|r|^2 \leq \alpha(1-d)$, $ad \geq \Theta_N$, and $\alpha, \eta_N^2 \leq e^{-2q\psi(s_N) + O_{h,\varpi}(k \log q)}$, the error in NV16 has this same doubled exponential rate. It requires no lower bound on μ .

There is a sharper direct receiver from the measured support itself:

$$\begin{aligned} \rho_f &= \frac{\mathfrak{s} - a|c_g|^2}{d}, \\ \max\{0, (\mathfrak{s} - a\eta_N^2)/d\} &\leq \rho_f \leq \min\{1, \mathfrak{s}/d\}. \end{aligned} \tag{NV17}$$

The identity follows by adding the two boundary-column squared overlaps in the definition of $\mathfrak{s}$. It improves a mass-free correction bound even when the terminal support is very small.

The native source parameter has its own proved growth. Put $\vartheta = E_N/\omega_N$, $\epsilon = \sqrt{E_N F_N}/\omega_N$, and $r_\omega = \omega_{N+1}/\omega_N$. Then

$$\beta = \frac{\epsilon^2 d}{\vartheta r_\omega}.$$

OCP5 gives $\vartheta \geq 1/2$ eventually and $\vartheta \leq 1$; RC25 gives $d \geq \Theta_N$, $\log \epsilon = q\psi(s_N) + O(k \log q)$, and $-\log \Theta_N = O(k \log q)$. The full source comparison in RC22-25 gives

$$\frac{\ell_k}{u_k}(N+1)(N+1/2) \leq r_\omega \leq \frac{u_k}{\ell_k}(N+1)(N+1/2).$$

Taking logarithms therefore proves, at every original cutoff,

$$\boxed{\begin{aligned} \log \beta &= 2q\psi(s_N) + O_{h,\omega}(k \log q), \quad t_{\text{peak}} = \frac{1}{\beta + 2} \\ &= e^{-2q\psi(s_N) + O_{h,\omega}(k \log q)}. \end{aligned}} \quad (\text{NV18})$$

The variable t_{peak} is the original covariance-step fraction, not the arithmetic evolution time.

Set $\kappa_{\text{term}} = \beta \rho_f / (1 + \beta)$. NV16 yields the signed and quantitative terminal receiver

$$\begin{aligned} 0 \leq \mu - \kappa_{\text{term}} &= (\mu - \rho_f) + \frac{\rho_f}{1 + \beta} \\ &\leq E_f + \frac{1}{1 + \beta}. \end{aligned}$$

Thus

$$\boxed{\begin{aligned} \Phi(t_{\text{peak}}) - \frac{1}{2}(\Phi_0 + \Phi_1) &= \frac{Y_0 \Delta \tau}{4} \kappa_{\text{term}}, \\ \left| \Phi(t_{\text{peak}}) - \frac{1}{2}(\Phi_0 + \Phi_1) - \frac{Y_0 \Delta \tau}{4} \mu \right| & \\ &\leq \frac{Y_0 |\Delta \tau|}{4} \left(E_f + \frac{1}{1 + \beta} \right). \end{aligned}} \quad (\text{NV19})$$

At the peak the weighted endpoint chord is exactly the midpoint, by NV9. The coefficient error in parentheses has the doubled exponential rate. If $\Delta \tau = 0$, both sides of the error bound are zero; division by the trace change is never required.

This also gives an absolute normalized-current error on the two original adjacent paths. At each endpoint the exact trace is $\tau_N = -2\epsilon_N \Im r_N$, so

$$\begin{aligned} |\Delta \tau| &\leq 2\epsilon_N \sqrt{\alpha_N} + 2\epsilon_{N+1} \sqrt{\alpha_{N+1}} \\ &\leq e^{O_{h,\omega}(k \log q)}. \end{aligned}$$

Multiplying this bound into NV19 shows that its error divided by Y_0 is at most $e^{-2q\psi(s_N) + O_{h,\omega}(k \log q)}$. Both endpoint leakages and both complete source metrics remain in the finite bound; no signed trace value has been evaluated.

The source step's quadratic coefficient receives the same mass estimate:

$$\boxed{\begin{aligned} q_2 &= -(1 + \beta)\beta Y_0 \rho_f \Delta \tau, \\ \left| \frac{q_2}{(1 + \beta)^2} + Y_0 \mu \Delta \tau \right| &\leq Y_0 |\Delta \tau| \\ &\left(E_f + \frac{1}{1 + \beta} \right). \end{aligned}} \quad (\text{NV20})$$

The scaling on the left is exactly the coefficient of λ^2 in NV9. This preserves the original SC17 coefficient and gives its native terminal-mass receiver rather than assuming a current-sector occupation.

5. The exact source-step sign cone

Write $A = \Phi_0$, $B = \Phi_1$ and $\kappa = w\Delta\tau/(1 + \beta)$. The same original current is

$$F(\lambda) = A(1 - \lambda) + B\lambda + \kappa\lambda(1 - \lambda).$$

It is nonnegative on the complete source step if and only if

$$\boxed{A \geq 0, \quad B \geq 0, \quad \kappa \geq -(\sqrt{A} + \sqrt{B})^2.} \quad (\text{NV21})$$

Necessity of the endpoint signs follows by evaluation. For the interior divide by the positive number $\lambda(1 - \lambda)$:

$$\frac{F(\lambda)}{\lambda(1 - \lambda)} = \frac{A}{\lambda} + \frac{B}{1 - \lambda} + \kappa.$$

For $A, B > 0$, completing a square gives

$$\begin{aligned} & \frac{A}{\lambda} + \frac{B}{1 - \lambda} - (\sqrt{A} + \sqrt{B})^2 \\ &= \frac{[\sqrt{A}(1 - \lambda) - \sqrt{B}\lambda]^2}{\lambda(1 - \lambda)}. \end{aligned}$$

Its infimum is $(\sqrt{A} + \sqrt{B})^2$, attained at $\lambda_* = \sqrt{A}/(\sqrt{A} + \sqrt{B})$. If $A = 0, B > 0$, the infimum is B , approached as $\lambda \downarrow 0$; the case $B = 0, A > 0$ is the opposite endpoint. If $A = B = 0$, it is zero identically. These cases prove both directions of NV21, including every zero-endpoint limit. Strict positivity on the closed step requires $A, B > 0$ and a strict third inequality.

Equivalently the exact coefficient matrix

$$\boxed{C_{\text{step}} = \begin{pmatrix} A & (A + B + \kappa)/2 \\ (A + B + \kappa)/2 & B \end{pmatrix}} \quad (\text{NV22})$$

is copositive: its quadratic form is nonnegative for every vector with nonnegative real coordinates. Indeed at $(x, y) = (1 - \lambda, \lambda)$ it equals $F(\lambda)$, and every nonzero nonnegative vector is a positive multiple of such a vector. This proves an exact map between the original covariance path and this two-dimensional cone, with no change to its source column or endpoint metrics.

When $A, B > 0$, the boundary $\kappa = -(\sqrt{A} + \sqrt{B})^2$ has

$$\begin{aligned} F(\lambda) &= [\sqrt{A}(1 - \lambda) - \sqrt{B}\lambda]^2, \\ \boxed{t_* &= \frac{\sqrt{A}}{\sqrt{A} + (1 + \beta)\sqrt{B}}.} \end{aligned} \quad (\text{NV23})$$

This is the actual source coordinate corresponding to λ_* . Below the boundary, evaluation at that same t_* is strictly negative. When one endpoint is zero, a violation of NV21 produces a negative value arbitrarily near the corresponding endpoint by the explicit interior formula above. Thus failure of the cone inequality identifies the exact sign-changing source paths rather than declaring the interpolation unrelated to the sign problem.

Applying NV21 to $-F$ gives the complete nonpositive cone:

$$\boxed{A \leq 0, \quad B \leq 0, \quad \kappa \leq (\sqrt{-A} + \sqrt{-B})^2.} \quad (\text{NV24})$$

In NV21–24 the coefficient is the original $\kappa = w\Delta\tau/(1+\beta)$. For the native terminal source it is also $Y_0\kappa_{\text{term}}\Delta\tau$, with the full doubled-rate mass estimate NV19. Hence these are exact necessary and sufficient endpoint-and-trace tests for the actual marked source path. They do not assert which cone contains a native path before its endpoint data and trace change are evaluated.

6. Adjacent dual-number fibres and native source curvature

At each of the two actual endpoints $i = N, N+1$, use that endpoint's own consecutive source vectors e_i, f_i , full source metric G_i , minimum isometry J_i , and observed columns $u_i = J_i^\dagger e_i$, $v_i = J_i^\dagger f_i$. Write

$$a_i = \|u_i\|^2, \quad d_i = \|v_i\|^2, \quad r_i = u_i^* v_i, \\ \mathfrak{d}_i = a_i d_i - |r_i|^2 \geq \Theta_i > 0.$$

Here a_i is source support retention, not either scalar endpoint value of the function ψ . For a complex deformation parameter σ , DF3–12 gives

$$F_{B,i}(\sigma) = (\epsilon_i v_i + \sigma u_i) u_i^*, \\ F_{B,i}(\sigma)^2 = (a_i \sigma + \epsilon_i r_i) F_{B,i}(\sigma).$$

Multiplication proves this identity directly. Its collision is $\sigma_{*,i} = -\epsilon_i r_i / a_i$. In the original orthonormal current frame, $F_{B,i}(\sigma_{*,i}) = \epsilon_i \sqrt{\mathfrak{d}_i} h_i g_i^*$. It is square-zero and nonzero because $\epsilon_i > 0$ and $\mathfrak{d}_i > 0$; positivity of a_i alone would not prove nonzero nilpotence. The support-plane algebra at that parameter is therefore faithfully the dual-number algebra, with its actual nonzero generator just displayed.

The selfadjoint part of the full endpoint action contributes zero trace current. Its rank-one boundary part gives $\tau_i = -2\epsilon_i \Im r_i = 2a_i \Im \sigma_{*,i}$. Define the weighted adjacent collision displacement $\chi_{\text{coll}} = a_{N+1} \Im \sigma_{*,N+1} - a_N \Im \sigma_{*,N}$. Then the exact original source receiver is

$$\Delta\tau = 2\chi_{\text{coll}}, \quad q_2 = -2(1+\beta)w \chi_{\text{coll}}. \quad (\text{NV25})$$

No endpoint is compared in the other's metric. This identity uses the two actual trace scalars, each computed from its own minimum section.

The native peak and the complete path curvature become

$$\Phi(t_{\text{peak}}) - \frac{1}{2}(\Phi_0 + \Phi_1) = \frac{Y_0 \kappa_{\text{term}}}{2} \chi_{\text{coll}}, \\ \frac{d^2 \Phi}{d\lambda^2} = -\frac{4w}{1+\beta} \chi_{\text{coll}}. \quad (\text{NV26})$$

Both formulas follow by substituting NV25 into NV9 and NV19. Thus the adjacent imaginary displacement of the dual-number fibre, with its actual support weights, controls the complete original marked-current curvature. NV19 retains the doubled-rate terminal-mass approximation to κ_{term} ; NV21–24 receive $\kappa = 2w\chi_{\text{coll}}/(1+\beta)$ without a sign assumption. The complex deformation parameter σ , covariance coordinate t , and its exact parameter map $\lambda(t)$ remain distinct and explicitly related by these equations.

7. Source scope and checks

The full SC1–25 proof and ST1–20 proof accompany the current local edition. Both links use the verified public support-transport edition pin; their complete source dependencies remain retained with that edition. NV1–11 rederive all finite interpolation statements directly, including their minimum maps and edge cases. NV12–20 use the

actual native source column and proved original source/terminal estimates, with the public predecessor locators given above.

The checker constructs covariance sums from actual consecutive monic Gamma polynomials in an auxiliary four-root quotient, with its complete relations and cutoff minima. It verifies the native recurrence, unmetricized skew current, every interpolation and trace coefficient, and the moving minimum maps exactly. These finite matrices test the formulas; they are not observations of a hypothetical zeta zero or replacements for the original five-orbit map in the proof. Separate exact complex two-column checks test NV16's phase and the support-mass receiver. No numerical evidence is used to select an actual terminal sign.